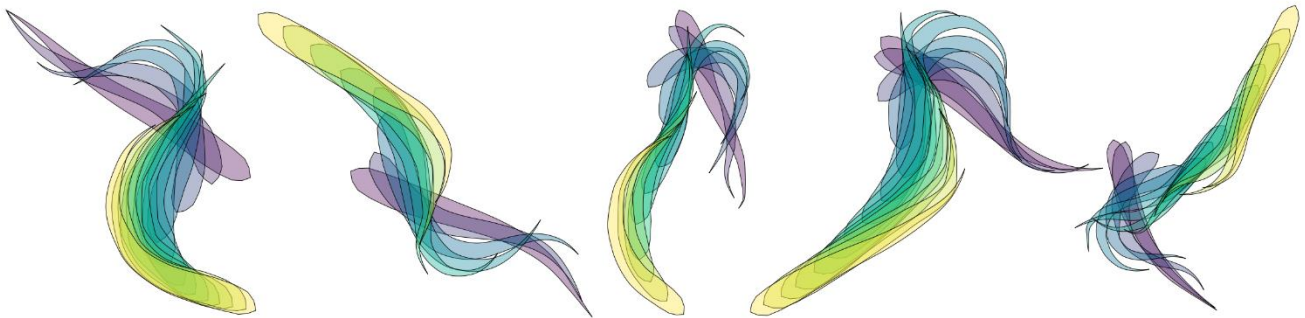


Effect of hydrological events on the steady/unsteady swimming in Salmonidae (*Salmo trutta*)



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Master 2 Biodiversity Ecology Evolution « Biodiversity and Tropical Ecosystems
Mention: Aquatic, Littoral and Insular Ecosystems (BEST-ALI) »

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Milesker Anitz!

Eskerrik asko guztioi!

Abstract:

Morphology is a major physical constraint on fish locomotion. Swimmers morphotypes are described because of their specialisation in distinct environment with different swimming capacities. Assessing correct differences in fish performances can become a real challenge when observed at the micro-evolutive scale with similar morphology. We tried to evaluate possible differences in the steady/unsteady swimming of the species *Salmo trutta* facing different flow regimes conditions. As floods are predicted to increase in term of intensity and frequency, we tried to detect their impact through animal experimentation. One group faced fluctuant flow regime while the other stayed in stable conditions. We collected data on steady swimming against a constant flow, and unsteady swimming mixing behaviour, kinematics and induced escape reactions (fast-start). We measured performances and behaviour metrics to evaluate the best potential in detecting differences across treatment. Genetic lineage was also considered (reared vs. wild origin). Statistical analysis and modelling showed no significant metric to distinct clearly between origin nor treatment for the steady swimming and the behaviour analysis. However, statistical modelling showed high significant importance of genetic origin in fast-start actuation across time suggesting better performances of the wild trout in the early stage of fast start. Treatment remained undetected as a discriminative factor in all analysis suggesting that the treatment was not intense enough or that the experimental conditions allowed individuals to find shelters in the semi-natural environment. This could also indicate high robustness of the developmental patterns. Results could be valuable to pre-evaluate repopulations efficiency of Salmonidae populations in river ecosystems.

Keywords: Morpho-functionality, Swimming performances/kinematics, Steady/unsteady performances, *Salmo trutta*, Micro-evolution

Résumé :

On distingue plusieurs morphotypes adaptés à des environnements distincts. Evaluer les différences de performance entre les individus peut s'avérer complexe à l'échelle micro-évolutive. Nous avons cherché à évaluer les différences potentielles dans la nage entre stabilité et manoeuvrabilité chez la truite commune (*Salmo trutta*) soumise à différents régimes hydrologiques. Face aux augmentations de l'intensité et de la fréquence des crues, nous avons testé leur impact via l'expérimentation animale. Un groupe est soumis à un régime hydrologique fluctuant, tandis que l'autre est maintenu en conditions stables. Des données sont recueillies sur les performances de stabilité, et de manoeuvrabilité, incluant comportement et réactions de fuite. Plusieurs paramètres de performance et de comportement ont été mesurés afin d'évaluer leur potentiel de détection des différences entre les traitements. L'origine génétique a été considérée (élevage vs. sauvage). L'analyse et la modélisation statistique n'ont révélé aucun paramètre suffisamment significatif pour distinguer clairement le traitement parmi les individus pour l'analyse de stabilité/manoeuvrabilité et pour l'analyse comportementale. Cependant, la modélisation statistique a révélé une importance significative de l'origine génétique dans les réponses de fuite, suggérant de meilleures performances des truites sauvages lors de la phase initiale du fast-start. Le traitement n'a pas été identifié comme un facteur discriminant dans toutes les analyses, ce qui laisse penser qu'il était insuffisant ou que les conditions expérimentales permettaient aux individus de trouver refuge dans le milieu semi-naturel. Ces résultats pourraient s'avérer pertinents pour pré-évaluer l'efficacité des efforts de repeuplements concernant les populations de Salmonidae dans les rivières.

Mots-clés : Morpho-fonctionnalité, Compromis de nage stabilité/manoeuvrabilité, Performances/cinématiques de nage, Micro-evolutive scale, *Salmo trutta*

Glossary:

Fast-start: Rapid swim burst during prey-predator interaction. Mechanical reflex directly linked to the individual performance and capacity.

Redds: shallow, gravel-bottomed patches of streams arranged by female trout for laying eggs. Allows better oxygenation and protection for the eggs.

Bief (reach): section of the semi-natural flume enclosed with two mesh allowing water flow. The flume was divided in two sections upstream/downstream divided into 6 biefs each.

CM: Centre of Mass of the individual. Point where the forces apply during fast start event. Also represents the centre of gravity of the animal.

Thigmotaxis: Behavioral response where animals, when placed into a new enclosed space, tend to stay near the perimeter while avoiding the interior area.

SGM: Segment Growing Method algorithm developed by [Fetherstonhaugh et al., 2021](#), improved and provided by [Sterling R.](#) and [Akanyeti O.](#)

Random Forest algorithm: Decisional Tree - based algorithm to discriminate explicative variables using bagging method with independent decisions.

GAM/GAMM: General Additive (Mixed) Model for statistical modelling of non-linear relationship with the potential addition of random effects.

REML: Restricted Maximum Likelihood, estimation method for variance components of statistical models that less biased estimates than Maximum Likelihood in general.

AIC: Akaike Information Criterion is an estimator of prediction error for statistical model. It is calculated from the Likelihood (L) of the model and the number of parameters (k):
 $AIC = -2\log(L) + 2k$.

Edf: Effective Degrees of Freedom estimate the complexity of spline fitted for statistical model and informs us about the correct estimation of parameters (edf~k).

IC95%: Confidence Interval of a value estimated with a 5% first order risk.

GLS: Generalised Least Squares method used to estimate the unknown parameters in a linear regression model.

PLS: Partial Least Squares method used to estimate the unknown parameters in a linear regression model.

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1. Introduction

Turbulent flow is a major pressure on the physiological development of fish larvae. Through environmental conditions and natural selection, evolution has led to the emergence of different morphotypes in terms of swimming performances (Blake, 2004; Satterfield *et al.*, 2023). Riverine species face varying conditions of flow exposure during their life cycle. Exposure to high varying environment due to global change can induce distinct morphological development and leads to differences in swimming performance (Langerhans, 2008; Oufiero & Whitlow, 2016).

1.1 Global change and hydrological events

As global warming is rising in the Anthropocene, so are climatic events (e.g. droughts, floods). These natural events can cause and induce important changes in ecosystems and threaten wildlife, ecosystems, and even civilizations (IPCC, 2001; IPCC, 2023). Floods and droughts in particular are natural hydrological events driven by rainfall and climatic variations that are part of the natural flow regime of rivers and regulate river ecosystems (Parasiewicz *et al.*, 2019).

Hydrological systems are highly influenced by climatic conditions. Events such as droughts, flood are observed naturally in rivers, but those events are becoming more frequent and with higher intensity due to global change (Lehner *et al.*, 2006; Dankers & Feyen, 2008). Floods are described as ‘rise, usually brief, in the water level of a stream or water body to a peak from which the water level recedes at a slower rate’ (WMO, 2012). Typical 100-years floods are projected to be more frequent and more intense in large areas of southern and south-eastern Europe (Lehner *et al.*, 2006). Talbot *et al.* (2008) propose a review of studies that list negative and positive impacts of floods on ecosystem services finding that small floods could enhance many services, where extreme floods are associated with declines of the mentioned services. Despite floods being a natural component of the river flow regime, more frequent extreme floods could dangerously impact the ecosystem recovery capacity (Swanson *et al.*, 1998; Talbot *et al.*, 2018). River ecosystems are directly and indirectly affected by floods. It can cause landslides (Talbot *et al.*, 2018), habitat destruction, and affect the quality of water, deeply modifying environmental abiotic conditions (Hrdinka *et al.*, 2012). Droughts, which are hydrological events, are different from low flow, which is a seasonal phenomenon defined as ‘flow of water in a stream during prolonged dry weather’ (Smakhtin, 2001; WMO, 2012). Droughts are mostly driven by climate fluctuations and are characterized by occurrence, duration, deficit volume (Tallaksen, 2000) and result in a lack of

water availability. It can also be influenced by human activity ([Smakhtin, 2001](#)). Their impact on river ecosystems can be from marginal to permanent changes to ecosystems shifts and extinctions ([Parasiewicz et al., 2019](#)). However, the expected general consequence of long-term natural floods and droughts is an increase in biological diversity and ecosystem health in rivers as these act as rejuvenating events that maintain variability within the ecosystem ([Parasiewicz et al., 2019](#)).

“ Understanding functional relationships between flow patterns and biological consequences is of the utmost importance for planning adaptation measures to climate change and for sustainable river management “ ([Parasiewicz et al., 2019](#))

1.2 River ecosystems

Freshwater environments represent less than 1% of the Earth's surface and host 12% of global non-microbial species ([FWBON, 2022](#)). This biodiversity represents almost one third of the vertebrates, almost all of the amphibians and half of the fishes ([FWBON, 2022](#)). However, they are among the most threatened ecosystems on earth ([Hoppenreijns et al., 2024](#)). Rivers are inland-water/freshwater ecosystems with natural flow regimes. Rivers in temperate climate exhibit seasonal variability in terms of flow regime, temperature, and presence of water ([Davies and Walker, 2013](#)). During summer, it is common to find little or no water in some parts, whereas in the winter, the flow rate is generally much stronger. The intensification of droughts and floods with higher frequency can cause problems for the capacity of species to maintain in those ecosystems ([Leigh et al., 2015](#)). Species that are specialists with low phenotypic plasticity present lower resistance and resilience against frequent variations of the systems ([Thompson, 1991](#); [Price et al., 2003](#); [Sánchez-González & Nicleza 2017](#)). Depending on the intensity of the event, the damage can differ greatly. Absence of water due to droughts forces aquatic species to certain zones, or prevents them from emigrating, therefore limiting reproduction, dispersion, resources availability. As well, floods with strong flow stop species that can't swim through high flow to the same constraints. Floods also cause higher erosion on the physical systems and can move land, producing obstacles and destroying habitats. Anadrome/Lacustrine fishes, in their life-cycle, are known to go up in rivers to reach specific places in order to reproduce and lay down their eggs. Extreme events can prevent their capacity to overcome habitual obstacles, especially high flow. As turbulence rate often increases with higher flow, energy expenditure (cost of swimming)

increases as well (Hinch & Rand, 1998; Enders *et al.*, 2003). Overcoming higher flows forces them to use more energy than usual or constrain them to change reproduction zones, and therefore the development zone.

1.3 Brown trout as a biological model

Trout is a common biological model used to study migration of anadrome species (Masson *et al.*, 2018), bio-invasion of introduced species (McIntosh *et al.*, 2012; Vignon *et al.*, 2023; GISD, 2026), fish kinematics and adaptive response to hydrological events (Domenici & Blake, 1997; Liao, 2007; Kerr *et al.*, 2016; Li *et al.*, 2021). The Brown trout *Salmo trutta* Linnaeus, 1758 is a Salmonidae commonly found widespread across the world (Fig. 1) with high adaptive capacity. It is currently classed as a Least Concern (LC) species (IUCN, 2026).

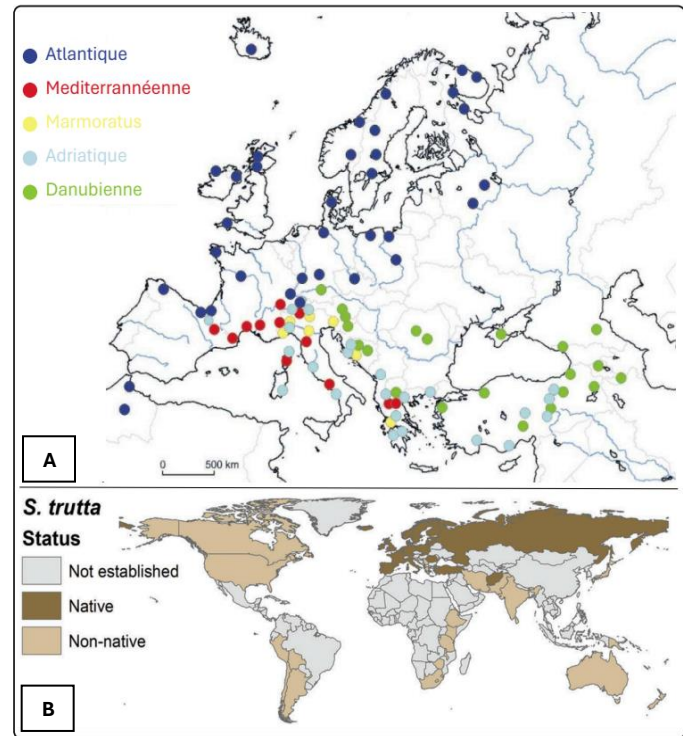


Figure 1: **A)** Map of the distribution of *S. trutta* across the world with native and non-native (introduced) regions highlighted. Adapted from Muhlfeld *et al.*, 2019. **B)** European spatial distribution of the five genetic lineages of *S. trutta*. Adapted from Bernatchez, 2001. In this paper, trout from Mediterranean and Atlantic lineages are studied.

The life cycle of the brown trout (Fig. 2) consists mainly of freshwater phases. Brown trout are iteropare, once they become mature, adults reproduce once a year, several times (Rochard & Elie, 1994). Spawning takes place in outlets or rivers in shallow freshwater with enough oxygenation. The female covers the eggs by restirring the sand and fine gravel into redds* (Kailola *et al.*, 1993). After hatching at 12mm, larval brown trout remain in the gravel for 2-3 weeks (depending on the °C/day conditions) until they are about 25mm long, when they emerge to begin feeding in the water column. Brown trout is a species with high philopatry, meaning they are likely to come back and reproduce where they originally hatched (Rochard and Elie, 1994). There are two forms of brown trout: resident and anadrome (Masson *et al.*, 2018). Resident forms are individuals living mostly in lakes and going up and down inlets/outlets to reproduce and develop.

Anadrome form includes a salt-water phase. Juveniles spend 1-5 years in fresh water before descending rivers to access sea and salt water.

Smoltification is the transition phase where juveniles move close to shore and estuaries not far from their native rivers. Individuals can stay in salt-water for 6 months to 5 years (Jonsson, 1985; Rochard & Elie, 1994; Fishbase, 2026). The resident form is the one we focused on in this study.

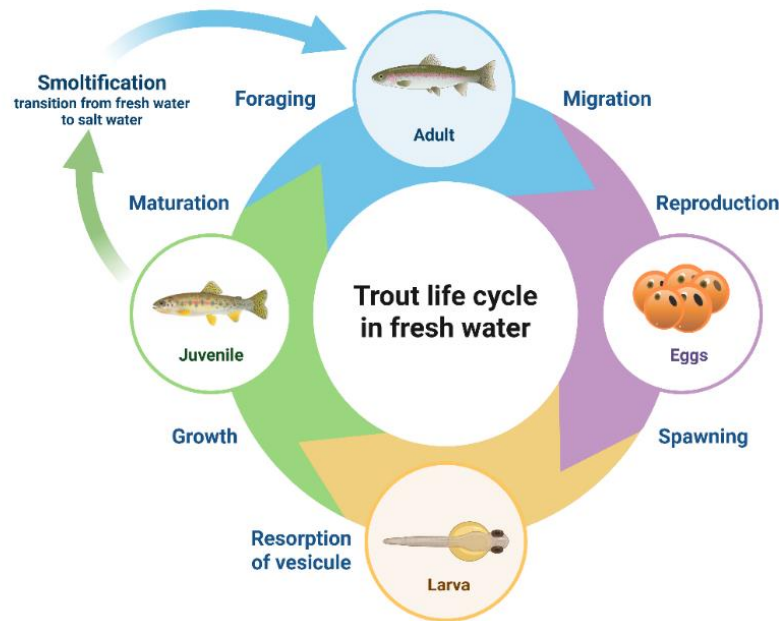


Figure 2: Life cycle of the brown trout *Salmo trutta* with different phases. *S. trutta* is a riverine species with high philopatry. Description of life cycle based and adapted from (Jonsson, 1985; Rochard & Elie, 1994; Fishbase, 2026). Created using BioRender.

Studying freshwater organisms like *S. trutta* and its response to varying environments has direct implications for conservation of freshwater organisms (Shadwick and Lauder, 2006; Castro-Santos and Haro, 2006; Muhlfeld *et al.*, 2019). In addition, trout and Salmonidae are used for pisciculture, recreation and represent a real economic issue (Holmlund and Hammer, 1999; Muhlfeld *et al.*, 2019). Salmonidae are also great indicators of biodiversity with adaptive capacity spread widely around the globe (Fig. 1A) (Bernatchez, 2001; Muhlfeld *et al.*, 2019).

1.4 Swimmers' morphs and functional trades-off

Teleost fish have evolved differently regarding the environmental conditions they faced during their life period (e.g. diet, habitat shift, temperatures) (Burns and Sidlauskas, 2019). At the end of the Upper Cretaceous, the number of teleost families experienced a serious increase, especially in hotspot zones like coral reefs (Rabosky *et al.*, 2013; Price *et al.*, 2015; Bellwood *et al.*, 2017).

Currently, most research on the ecology and evolution of fish has focused on features of fish design that are easier to measure, rather than aspects of locomotor performance (Langerhans and Reznick, 2009). It is however performance and not design, that is directly linked to individual fitness. Therefore, different types of morphology have been classified across fish morphotypes depending on their morphs and differences in terms of performance in propulsion and manoeuvring (Satterfield et al., 2023).

About performance, Webb (1984) classed fishes into three main categories: body and caudal fin (BCF) periodic, **BCF transient** (fast-starts, turns) and median and paired fin (**MPF**) swimmers (Webb, 1984; Blake, 2004). Therefore, specializing in one category could limit performance in the others. Trout, for example, is categorized as a generalist (Blake, 2004) resulting in a particular performance trade-off with no strong specialization in one category, with moderate performances in both stability and manoeuvring. About undulatory swimming specifically, several swimming modes are described depending on fish's kinematics and morphology (Breder, 1926; Lindsey, 1978). Most fishes swim primarily through undulation of their body, and caudal fin (BCF) (Di Santo et al., 2021) and four of the different swimming modes using caudal undulation/oscillation are described in the work of Di Santo et al., 2021 based on length of the propulsive body wave and the ratio of head-tail oscillation amplitude: anguilliform, subcarangiform, carangiform, and thunniform (Fig. 3, Appendix 1).

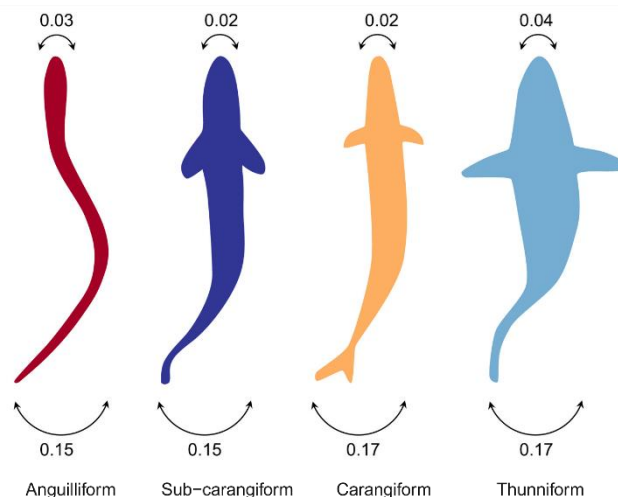


Figure 3: Analysis of steady swimming kinematics to illustrate the considerable similarity among the four classic fish swimming modes. Head and tail amplitude variables do not distinguish the four classic categories of BCF swimmers: eel (*Anguilla rostrata*), trout (*Salvelinus fontinalis*), mackerel (*Scomber scombrus*), and tuna (*Thunnus albacares*). Modified from Di Santo et al., 2021.

Subcarangiform are among others represented by the Salmonidae family, including salmon, trout, and others. Their swimming pattern is characterized by the action of both the head and tail, whereas anguilliform undulate the majority of their body and carangiforms and thuniforms reduce at their maximum the movement of the body, using only the tail (Breder, 1926; Lindsey, 1978; Di Santo *et al.*, 2021).

The performance trade-off results of natural selection are mostly driven by competition and predation (Langerhans, 2009). The most important trade-off in fish design and locomotion concerns steady and unsteady locomotion, the two primary swimming modes (Webb, 1984; Blake, 2004; Langerhans, 2008, 2009; Langerhans and Reznick, 2009). **Steady** swimming describes locomotion at constant speed in a straight line, and is commonly employed in nature during competition for limited resources (e.g. food, mating, favourable abiotic conditions) (Shadwick and Lauder, 2006; Webb, 2006; Langerhans, 2009). **Unsteady** swimming refers to more complicated locomotor patterns in which changes in velocity or direction occur, such as **fast-starts**, rapid turns, braking induced commonly by predator-prey interactions and social interactions (Walker *et al.*, 2005; Shadwick and Lauder, 2006; Webb, 2006; Langerhans, 2009; Li *et al.*, 2023). In stable conditions (without flow), fishes are likely to swim, or stay stationary, close to obstacles such as walls. This behaviour is called thigmotaxis* and is common when animals, including trout, tend to optimize space use or when stressed (Kerr *et al.*, 2016; Barany *et al.*, 2025). In addition, fast-starts are well-studied like the *Mauthner-cell* initiated escape response, called a '**C-start**' (Domenici and Blake, 1997; Blake, 2004; Domenici *et al.*, 2008; Langerhans, 2009; Vignon and Aymes, 2020). A fast-start is a high-energy swimming burst (Fig. 4) (Domenici and Blake, 1997). It represents a critical capability that significantly affects the consequences of hunting prey or escaping from predators (Fish and Lauder, 2006; Li *et al.*, 2023). Fast-start acts like mechanic reflex when a fish detects immediate predator like-danger or when a predator (e.g. birds, fish) try to strike the prey. The duration of a fast-start is around 30 ms for teleost fish (Li *et al.*, 2023), and the latency response to the stimulus is about 5 to 10 ms (Shadwick and Lauder, 2006; Wakeling, 2006). Because fast-start is a very brief mechanical response to stimulus, there is no high cognitive process involved. Indeed, it shows faster neuro-transmission than the common body bending wave

(Domenici and Blake, 1997; Li *et al.*, 2023) and allows studies to assess only the morpho-functionality of fast-start, neglecting the cognitive process.

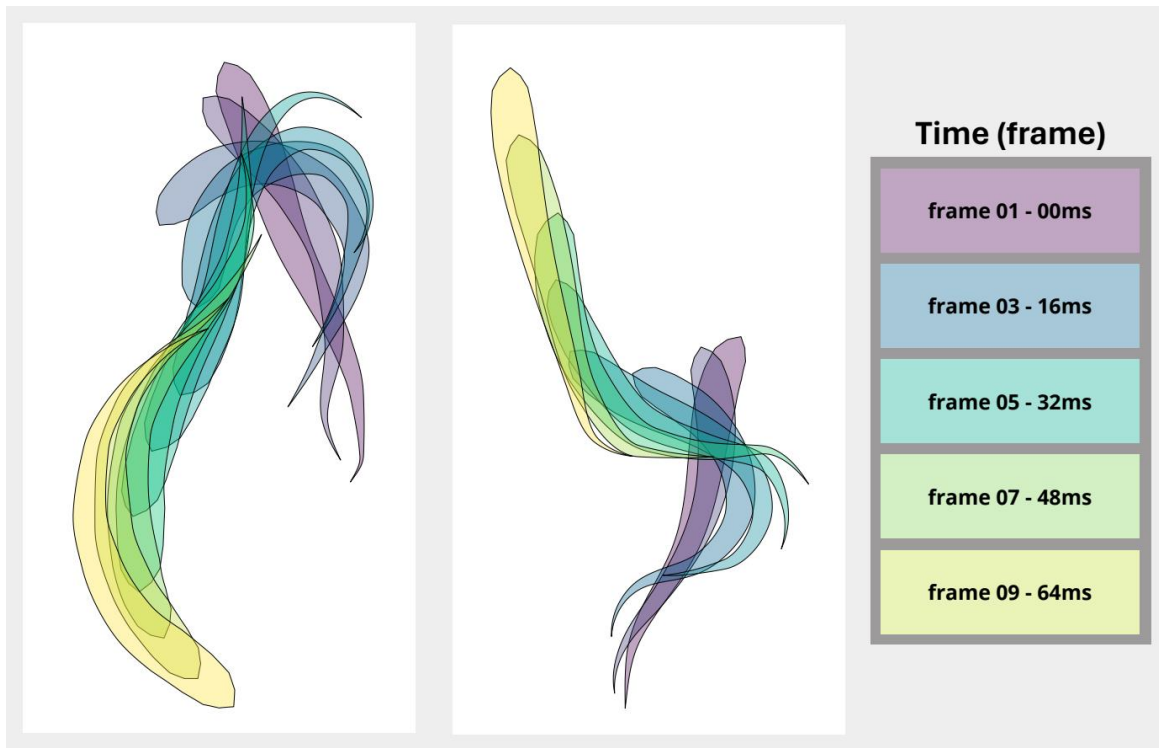


Figure 4: Illustration of two examples of a fast-start actuation through time. Body silhouettes are recreated using R from fast-start recording in 120 fps during unsteady swimming test.

Fast-start therefore enhance fitness in high-predation environments (Walker *et al.*, 2005; Langerhans, 2009). This controversy is well covered in the literature: in low-predation environment, natural selection should favour steady swimming with a lower cost of energy expenditure and higher competitive capacities; and high-predation environment favour unsteady swimming with an increase of escape and survival capacities (Katzir and Camhi, 1993; O'Steen *et al.*, 2002; Blake, 2004; Walker *et al.*, 2005; Langerhans, 2009). This idea states that predation is an important factor in the selection of functional traits in fish locomotion. It has been studied that fishes cannot optimize both swimming modes regarding propulsive mechanism issues that lead to two distinct phenotypes (Webb, 1984; Shadwick and Lauder, 2006; Wakeling, 2006; Langerhans, 2009).

Consequently, in a low-predation environment that should favour steady swimming, we expect that extreme hydrological variations could severely impact the steady/unsteady swimming capacities of fishes. We can expect different results **i)** in a non-constant flow with high varying intensity and frequency, individuals should show lower performance in steady swimming, and

maybe higher performances in manoeuvring and during fast-start events. **ii)** in opposition, strong flow could favour steady swimming performances at the detriment of unsteady swimming capacities.

However, this trade-off and swimming performances in general are mainly addressed at a multi-species scale with important morphological differentiations or distinct origin (Langerhans and Reznick, 2009; Di Santo *et al.*, 2021), giving pertinent information at a macro-evolution scale but little information on intra-specific variation. Studies do address the questions of morphological variation, across populations of the same species with divergence or plasticity (Hankison *et al.*, 2006; Langerhans, 2009; Bower *et al.*, 2018) and convergence in similar habitats (Franssen *et al.*, 2013). Genetic divergence and phenotypic plasticity are the main processes involved (Nei, 2007; Langerhans, 2008; Bower *et al.*, 2018). There is a need for knowledge about intra-specific morphological differentiation across varying flow and steady/unsteady performance in general (Langerhans, 2009). Morphological traits are direct responses to the environmental constraints (Bower *et al.*, 2018) and the performances of fishes underline their capacity to move, resist, and survive in varying environments. Selective regimes, such as fluctuant flow regime with extreme hydrological events, can generate and maintain phenotypic diversification (Hankison *et al.*, 2006).

In this study, we are investigating how the increase of **variation in flow regime** (due to the intensity and frequency increase of hydrological events) during the development of the brown trout *S. trutta* affects **intra-specific variation** of the swimming performances (stability and manoeuvring) and the general swimming behaviour in artificial conditions without flow.

2. Material and Method

2.1 Site and experimental setup

The brown trout lacustrine form is abundant in the French Nive river, an affluent of the Adour river, which runs for more than 300 km from the Pyrenees to the inner Bay of Biscay (Masson *et al.*, 2018). The experimental set-up was designed and conducted by M. Descat as part of the INRAE-team (Appendix 2). Individuals either faced fluctuant conditions (upstream) or stable conditions (downstream) and were exposed for 3 months (Fig. 5).

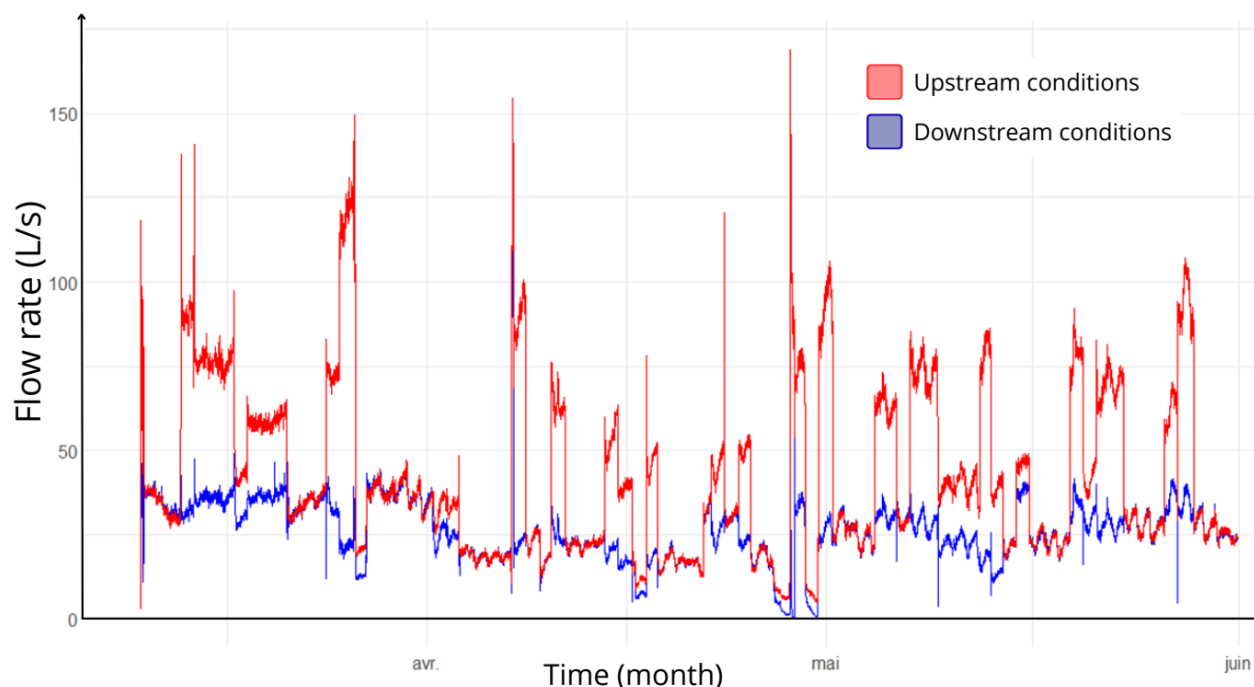


Figure 5: Illustration of the variation of flow rate during the 3 months of exposure to treatment inside the artificial river with fluctuant conditions (upstream) in red and stable conditions in blue (downstream).

2.2 Individuals

To examine the effect of the treatment applied to individuals regardless of the genetic origin, we collected eggs from two genetic groups, being the two most abundant genetic lineages around the south of France ([Fig. 1B](#)). Wild trout from the Mediterranean side (Cauterets) were collected by FDAAPPMA 65 and eggs were collected for our experience. The second batch of eggs came from pisciculture trout from an Atlantic genetic origin (Lees-Athas) ([Masson et al., 2018](#)). They should show similar development in morphology, although reared individuals generally show a longer phase of development ([Bohlin et al., 2002](#)) and higher competition. We expect reared trout to show higher performances with a ‘better’ development in a stable environment because of their better competitiveness in unaltered conditions. This serves two purposes: testing inter-lineage variation in terms of performance and the intra-lineage variation, between the two groups. Larvae from Lees-Athas were marked by immersion in Alizarin Red S solution (100 mg.L^{-1} , 3 hours) for later identification ([Vignon, 2018](#); [Caudron and Champigneulle, 2006](#)). When 80% resorption of the vesicle was achieved, individuals were inserted into an open small container with gravel in the experimental facilities of the Lapitxuri artificial river ([INRAE, 1980](#)), following the experimental protocol ([Appendix 2](#)). This allowed larvae to acclimate before moving through the water column with no action from experimenters.

2.3 Steady swimming analysis

Steady swimming pattern has been acquired by a team of M2 students of the [DYNEA Master - UPPA](#) in a collaborative project supervised by M. Vignon. Students collected and annotated steady swimming sequences from the videos previously recorded by M. Descat during steady swimming experiments (Appendices [2](#) & [3](#)). Videos were recorded at 40 fps, 1640*922 resolution using custom Raspberry PiScope cameras for 20 minutes. Extracted sequences consisted of two to three tail-beats where fishes were staying in the same location while swimming against a constant flow (mean flow speed: 0.169 m /s). After extracting each frame of movements, the body midlines of fishes were digitised using [ImageJ 1.54g](#) (Rasband *et al.*) (Schneider *et al.*, 2012) and a custom macro designed by M. Vignon ([Appendix 4](#)). The macro allowed semi-automated acquisition of the coordinates (15 equally-distant points) of the body midlines from snout to tail. All of the extraction, acquisition, and analysis protocols are available on request via [GitHub](#) (GitHub Inc., 2024, open-source repository). The steady swimming kinematics of fishes are to be compared in terms of head/tail amplitude and frequency of tail-beat with performances using correlation to determine the possible impact of the treatment, in potential interaction with genetic lineage on the stability/manoeuvrability trade-off at a micro-evolutive scale. At the end, we collected $n_1=108$ Cauterets individuals, 60% who faced fluctuant flow conditions, and $n_2=101$ Lees-Athas whom 48% who faced fluctuant conditions.

2.4 Unsteady swimming analysis

2.4.1. Individuals recording

To analyse the unsteady behaviour of individuals, we used an experimental tank ([Appendix 5](#)) to record individuals for 5 minutes while they were undisturbed. We used the Noldus tracking software [EthoVision XT 18.0.1832](#) to automate the tracking of individuals. It requires users to i) calibrate the area to track from for each video ii) to set track parameters for each video as well iii) to launch each acquisition one video at a time ([Appendix 6](#)). The metrics we address in behavioural analysis are not meant to detect differences to the thousandth of a second as the fish behaviour without current is relatively slow or still, and useful for fast-start inducement “starting from rest” (Domenici and Blake, 1997). Fish are also known to adopt hovering behaviour and conserve energy when conditions allow it (Webb, 1984; Ellerby and Gerry, 2011). In the idea of capturing an important effect that could be measured as a functional trait, micro-variations over thousands of

seconds are not of first interest. We therefore determined that 25 fps (lower than 120 fps initially recorded) was enough to track the behaviour pattern based on Benne, 2025 work, enough to detect stereotypical behaviours, or pattern differences, and to reduce the time of acquisition and energy consumption. EthoVision is used as a tracking software in academic and industrial use, by contrasting organisms with the background (Noldus *et al.*, 2001; Lim *et al.*, 2023). It is able to acquire tracks of three points designated as the centre of mass (CM*), the snout and the tail. It can provide the raw coordinates of each point for each frame and further analyse the speed, the cumulative distance, the rotational angle between the three points, acceleration, and time spent in specific zones. In this idea, we designated a zonation of the main area with three different zones: central zone, transition zone, and external zone (Appendix 6). We expect fish to spend most time in the external zone as thigmotaxis could be high with stressful conditions. Zones were approximately 2/3 of the fish's total length to allow fast-start kinematics inside the tank.

As they were placed in the experimental tank (Appendix 5), stimuli to induce fast-start response were realised by observers tapping the unfixed plank, propagating both sound-wave and shock-wave through the water. Roche *et al.* (2023) already described a semi-automated system to induce fast-start in previous works. They indicate how to prevent visual and sound disturbance by the observer, and how to reproduce their design, which have been tested, using a PVC pipe with a weight being released by mechanical action (Roche *et al.*, 2023). However, this resulted in a total absence of response from several fish. It has been found that tapping on the unfixed plank still produces a mechano-acoustic stimulus, and is the best way to induce fast-start response in our experience. The acoustic nature of the stimulus does not affect the mechanism induction of the fast start differently, but the stimulation must remain the same because of the sensibility of the *Mauthner-cell* (Eaton *et al.*, 1977; Roche *et al.*, 2023) and for repeatability. Fish were stimulated for ten minutes. Experimenters waited thirty seconds between each stimulus to let the fish recover.

2.4.2. Digitalization

Body midlines of fishes are to be digitised for each frame to assess precise movement description. About the digitalization (point by point) of the midline, several methods were considered. EthoVision has been tested and was reliable to detect behaviour, movements, speed and other metrics, but only allowed three points to be tracked (see earlier). Although the objective here is to

assess precise movement description and decomposition using more than three points during fast-starts events. Studies have already used three points to describe distance-related parameters and angular rotation (Vignon & Aymes, 2020) that have shown reliable results. However, we can find many publications on fish kinematics in biology and robotic designs improving body midlines digitalization to obtain more than three points (Tytell & Lauder, 2008; Fetherstonhaugh *et al.*, 2021; Di Santo *et al.*, 2021; Akanyeti *et al.*, 2022; Gong *et al.*, 2025). The main tools used are semi-automated MATLAB (MathWorks) custom software. After contacting Akanyeti O. (Di Santo's colleague), he redirected me into a previous work whose data, scripts, and publication were open access. We agreed that their MATLAB custom software CurveMapper4 (Appendix 7) developed by Lauder's team at Harvard was reliable to digitise body midlines into 200 equally distant points.

Their method was combined with a Python algorithm - Segment Growing Method (SGM) developed by Akanyeti's group (Fetherstonhaugh *et al.*, 2021). In order to have a realistic description of the fish curvature with minimal effort of modelling, five to six non-equal segments per fish are enough for steady swimming (Fetherstonhaugh *et al.*, 2021; Akanyeti *et al.*, 2022). Most parsimonious model revealed that the first segment, starting from the head, was the longest, correlating with (sub)carangiform swimming pattern (Fetherstonhaugh *et al.*, 2021). The objective is to use this method for our biological model to have a reliable decomposition of the trout kinematics during fast-start and to investigate unsteady behaviour using EthoVision (see-earlier).

2.4.3. Comparison of measurement method

In order to decide objectively which method was more relevant for our biological model, we tested the source of variation due to the methods ImageJ or MATLAB. Both methods have been used (Di Santo *et al.*, 2021; Vignon & Aymes, 2020) and were useful to different studies. We tested using Analysis of Variance (ANOVA) if the method was repeatable (find similar results by re-doing the same measurements) and replicable (different operators using the same method and find similar results). A small experiment was conducted to test variability across experimenters (for inter-operator variability) and also with replicates of the same input data (for intra-operator variability) that couldn't be distinguished from the intrinsic method variability. Nine fast-starts were randomly chosen to be measured. Three of them were measured by the four operators (two naive and two already familiar with those types of measures) with both methods resulting in $3 \times 4 \times 2 = 24$

measures in total. And the nine of them were measured 3 times by the same operator with the 2 methods, resulting in $9 \times 3 \times 2 = 54$ measures. Both methods were used to measure 200 points with custom scripts (Appendices 4, 7). We tested for differences in the relative distance to the mean point measured (Euclidian distance). To test the three potential causes of variance, the mean point was calculated **i)** with the 4 operators within each method (inter-operator variability) **ii)** with the 4 operators with the two methods (inter-method variability) **iii)** within the 3 replicates from the same operator within each method (intra-operator variability). The objective was to determine the proportion of variance explained by the fast-start event in opposition to the proportion explained by measurement bias. ANOVA between operators showed that the fast-start event was responsible for 52% of the total variance while the operator and method only explained $4 \times 10^{-6} \%$ and $2 \times 10^{-5} \%$ of the total variance, respectively (two-way ANOVA; Fast-start $F_{2,53654}=29564.18$, $p < 2.2 \times 10^{-16}$; Operator, $F_{3,53654}=0.15$, $p=0.93$; Method, $F_{1,53654}=2.06$, $p=0.15$). ANOVA between replicates of the same operator showed significant differences between the method but not between replicates (two-way ANOVA; Fast-start $F_{8,118933}=56168.37$, $p < 2.2 \times 10^{-16}$; Method, $F_{1,118933}=5.44$, $p=0.02$; Replicate, $F_{4,118933}=1.17$, $p=0.32$). However, the fast-start event explained 0.79% of the total variance, far greater than the replicates or the method with $8 \times 10^{-6} \%$ and $9 \times 10^{-6} \%$, respectively, of the total variance explained. Therefore, we could say with confidence that both methods were repeatable and replicable enough. Because the MATLAB software method was already published in open-access and is more practical to use than ImageJ, we chose it to digitise body midlines during fast-start events.

2.4.4. Metrics

To test for potential metrics that could help detect enough variability across treatment and genetic pool, we mainly separated metrics into two categories: time-related, and individual-related. First to analyse behaviour, EthoVision helps evaluating metrics such as the **frequency** and **time spent** in each of the three zones that could inform us on thigmotaxis, **mobility** as the percentage of mobility for the complete area of the detected animal that could inform us on the activity/stress of the animal, the **distance covered**, the **speed** and **acceleration**. In addition, we used R to calculate **head angle**, **tail angle**, and **curvature angle** differences across time (Fig. 6). To better characterize each individual, we used the variance of angles, acceleration, mobility, the maximum

of distance covered, the cumulated sum of angles differences, the time spent in each zone, the latency to joining the transition zone and the external zone, to have unique values per individual.

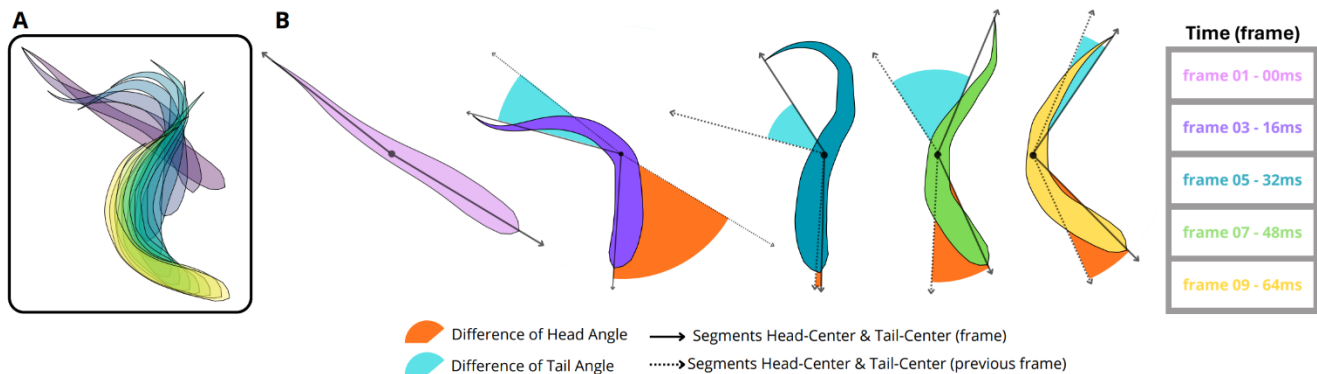


Figure 6: Illustration of the evolution of head and tail angle differences per frame during the fish swims (e.g. fast-start). **A)** Silhouettes of the fish during the 12 frames of the fast start with colour graduation. **B)** Examples of the difference in angle with 16ms lag during fast-start actuation.

As for the fast-start metrics, we exploit time-related metrics to detect differences in the short duration of the fast-start actuation. Studies were used to determine the CM position to calculate performance's metrics (Hale, 1996; Vignon and Aymes, 2020) (Appendix 8). To determine the CM position, where the forces apply, we were unable to weigh fishes as described in Roche *et al.* (2023) or use dorsal fin position like Vignon & Aymes (2020) which was hard to find on videos. Therefore, we used the definition of Webb (1978) that locates it at 38% of the total length on the rainbow trout, a close species to the brown trout. With the digitalization of the body midline, 38% consisted of the 76th of the 200 points from the nose. We used the CM to calculate **acceleration**, the **total distance covered** (distance cumulated) and the **efficiency**. The total covered distance is different from the effective distance (straight line between the CM initial and final position at the end of the sequence). We define the efficiency as the ratio between total distance covered and effective distance (Fig. 7). Total distance covered and acceleration are key-metrics in fast-start that inform on the probability of success of a prey-predator interaction reflecting the capacity to escape a predator (Blake, 2004; Walker *et al.*, 2005; Domenici *et al.*, 2008). Efficiency is a measure of the fish's efficiency to go from point A to B in a straight line. We calculated the angle of curvature twice in each frame, first with the whole half of

the body (head-centre, centre-tail), and secondly with the first 40 points and the last 40 points of the body midline, resulting in the **anterior-posterior ratio** and the **snout-tail ratio** ([Appendix 9](#)).

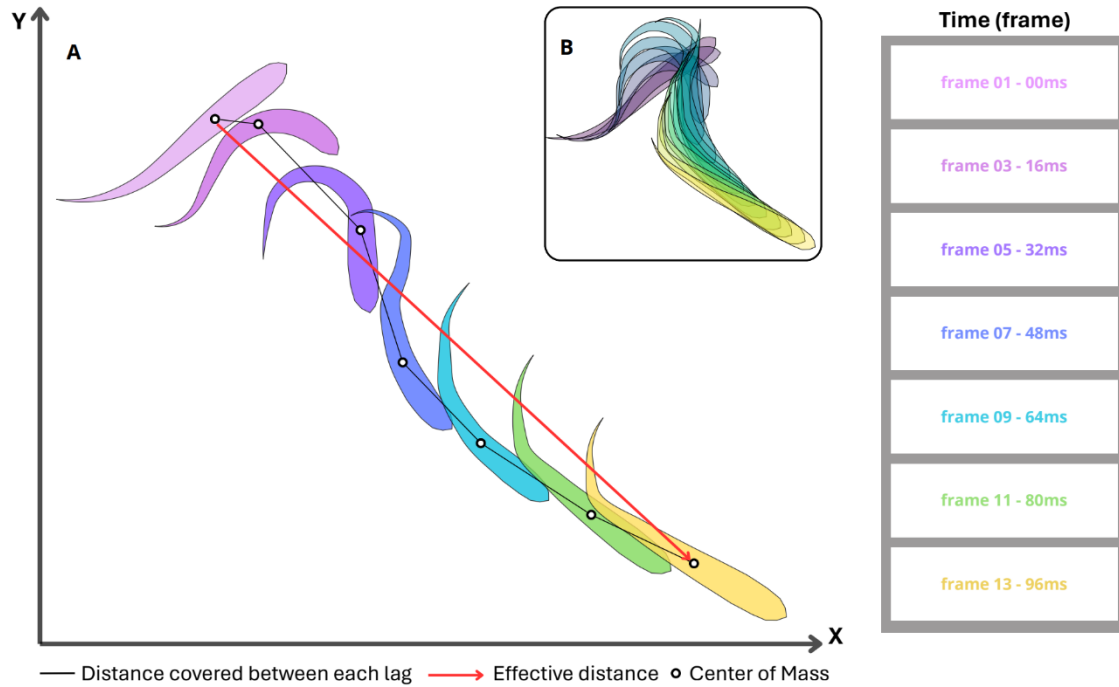


Figure 7: Illustration of the distance covered by the fish during a fast-start event with 16ms lag and the representation of the effective distance (red arrow). **A)** Fish actuation through space with colour graduation for time. **B)** Silhouettes of the fish during the 14 frames of the fast start.

In order to have a better and more realistic description of the fish kinematics, we used the SGM (see earlier) with Python to grow segments during fast-starts events. The secondary objective was also to use segments to detect if more variability came from the most extreme parts (snout and tail). Taking the maximum number of segments created per fish, we modelled 12 unique segments with R, with joint positions being the mean point positions of all fishes. We then calculated the **angle achieved by each segment** (abbreviated as 'segmentation') in each frame to detect if more angular variability came from different parts of the body midline. Finally, we calculated the cumulative sum of differences of all the angular variables to have time-related metrics and detect potential differences in temporal dynamics. **Latencies** before escape responses could not be measured here, as the stimulus was not visible in videos nor indicated. Those variables include information on the performances of fish to escape predators with power (acceleration, distance), agility (angles, curvature), and economy of movement (efficiency). They have been calculated for each frame of each fast-start, having different durations. Fast-start sequences were considered complete when fish started another turn, stopped moving, when the body midline was straight

again, or when the speed decreased enough for the fish to be considered almost still. It could be improved by the determination of phases 1 and 2, with the first and second bends, or with arbitrary time limits (Domenici and Blake, 1997; Walker *et al.*, 2005; Domenici *et al.*, 2008). All of the measured metrics are listed in (Appendix 10).

2.5 Statistical analysis

We explored individual-related metrics importance in the discrimination of treatment groups using Random Forest multivariate analysis (function “randomForest()” from package *randomForest* v4.7-1.2). The algorithm allows determining the relative importance of each variable in discriminating between either treatment or genetic origin. We set the tree size to 5000. Variable importance was assessed using the Mean Decrease Gini (MDG) corresponding to the average gain of purity by splits of a given variable, allowing the identification of the variables contributing most to group discrimination. We further examined the relative contribution and ranking of explanatory variables across the forest to identify the traits most strongly associated with treatment or genetic origin. To test for differences between individuals across groups, we used statistical modelling to fit the best model possible to our data. All the time-related metrics mentioned earlier were fitted along different General Additive (Mixed) Models to find the best predicting model that we could implement (functions “gam()” and “gamm()” from package *mgcv* v1.9-1). We used as explicative factors: the *treatment*, the *genetic origin*, and the *size*. The *weight* was highly correlated with *size* (Pearson correlation; $r = 0.959$; $t_{2426} = 166.54$; $p < 2.2 \times 10^{-16}$) so we decided to only use *size* because it could have more constraints on angular-related and distance-related metrics (Hale, 1996; Domenici and Blake, 1997). We also added the interaction between treatment and origin. We had to test the individual and fast-start event as a random effect that have high structural influence on the data. We transformed both ratios that were signed (negative values for left bend) into absolute values. We fitted GAMs and GAMMs with restricted maximum likelihood (REML). Time was included as a smooth term and candidate models differed according to the inclusion of treatment, genetic origin, size, localisation from upstream to downstream in the experimental river (bief*), the tank temperature during fast-start inducement and interaction-specific smooths. To discriminate candidate models, we used the Akaike Information Criteria (AIC) which is a direct measure of the model’s likelihood (L) penalized by the number of parameters (k), and we also calculated AIC weights (AICw). Statistical analysis, and

statistical models were implemented into [RStudio 2026.01.0](#) (R Core Team, 2024), and Python was used for SGM*, joint positions determination (see earlier) and animation.

3. Results

3.1 Exploratory analysis – Random Forest

An exploratory analysis has been realised on the variables calculated during behaviour recording using EthoVision and R. Conducted analysis did not return a high potential of discriminative measure for treatment conditions. The Random Forest (RF) analysis identified the two variables ‘Latency to transition zone’ and ‘Cumulated difference of head center angle’ as the behavioural variables contributing the most to treatment classification according to the MDA criterion ([Fig. 8](#)). The RF model (“Upstream” vs “Downstream”) had an average error of 48.2% indicating a low predictive performance in classifying the treatment suggesting that no single behavioural descriptor strongly discriminated individuals.

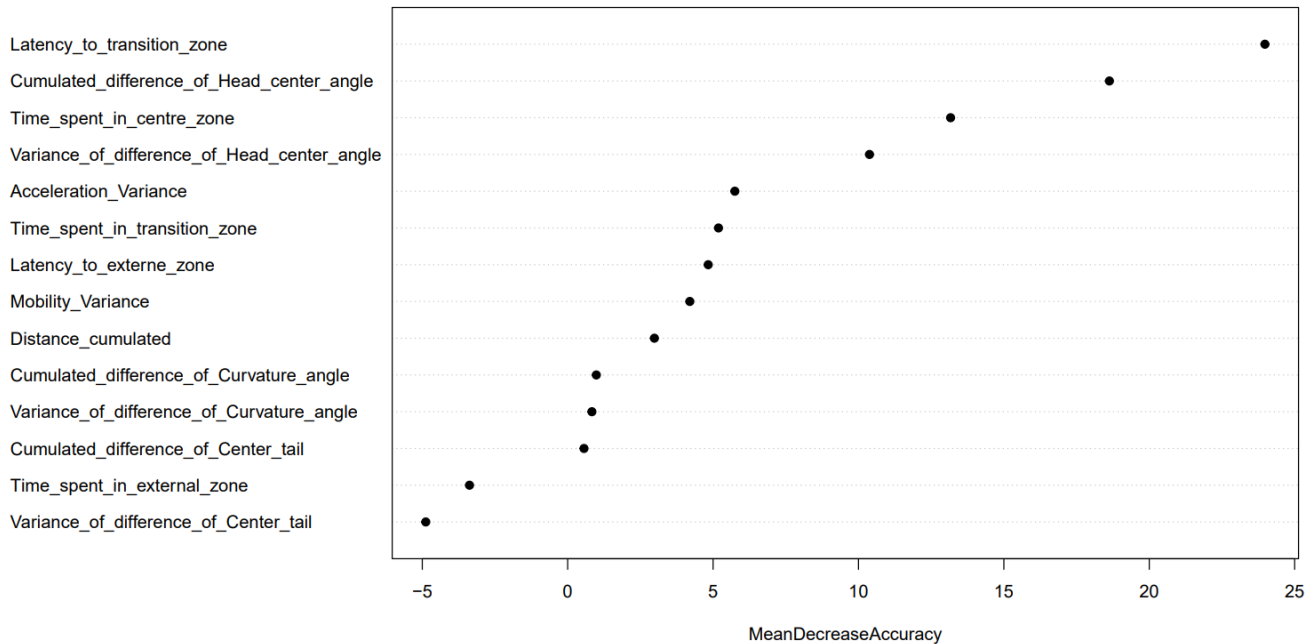


Figure 8: Relative importance in the Random Forest model in the discrimination of treatment groups along all of the metrics measured/calculated during behavioural test. Mean Decrease in Accuracy (MDA) shows the relative variable importance.

The first ranked variable: ‘*Latency to the transition zone*’ is a measure of the time before the fish started swimming and left the centre zone (starting zone), before exploring the arena. It was higher in the downstream group (median = 71.6s) than in the upstream group (median = 31.4s), although this difference was not statistically significant (Wilcoxon rank-sum test; $W = 2860$; $p = 0.088$; size

effect $r = 0.133$) and variances did not differ between groups (Levene's test; $F = 1.40$; $p = 0.24$). The second variable: '*Cumulated difference of Head-Center angle*' indicates potential differences in head's movement variability depending on the treatment. It showed a significant difference between treatments according to the (Wilcoxon rank-sum test; $W = 4059$; $p = 0.028$; size effect $r = 0.171$) and variances were homogeneous between groups (Levene's test; $F = 0.31$; $p = 0.58$). The upstream group showed a smaller mean difference in cumulated head movement (median = 21407 deg.) than in the downstream group (median = 24050 deg.). Random Forest classification suggests no strong descriptor of behavioural differences, with no strong relative importance between variables and high error of classification. A model was also fitted for genetic group classification sorting '*Acceleration Variance*' and '*Variance of difference of Curvature angle*' as the most explicative variables meaning that fishes could show differences mostly in curvature variability and acceleration. The model for genetic groups classification had an average error of 29% with no strong relative importance ([Appendix 11](#)). We tested genetic groups classification with treatment conditions to test interaction, and it resulted in a better discrimination for the individuals that underwent the varying flow regime (Upstream, error = 24%; Downstream, error = 31%) suggesting that the interaction between origin and treatment could be important and may have an influence on behavioural traits.

3.2 Statistical modelling – GAM and GAMM

We used statistical modelling to try to fit the best model to our data. The current models have been explored and fitted on five different time-related variables measured on fast-start events using REML method. We classified models along AIC and AIC weights to determine the best fit and kept the most parsimonious model for each response variable as depicted in ([Fig. 9](#)). To assess the robustness of model selection, all candidate models were also fitted using M method. As the ranking of models was consistent with that obtained using REML ([Appendix 12](#)), conclusions were unchanged. We therefore used parameter estimates from the REML models

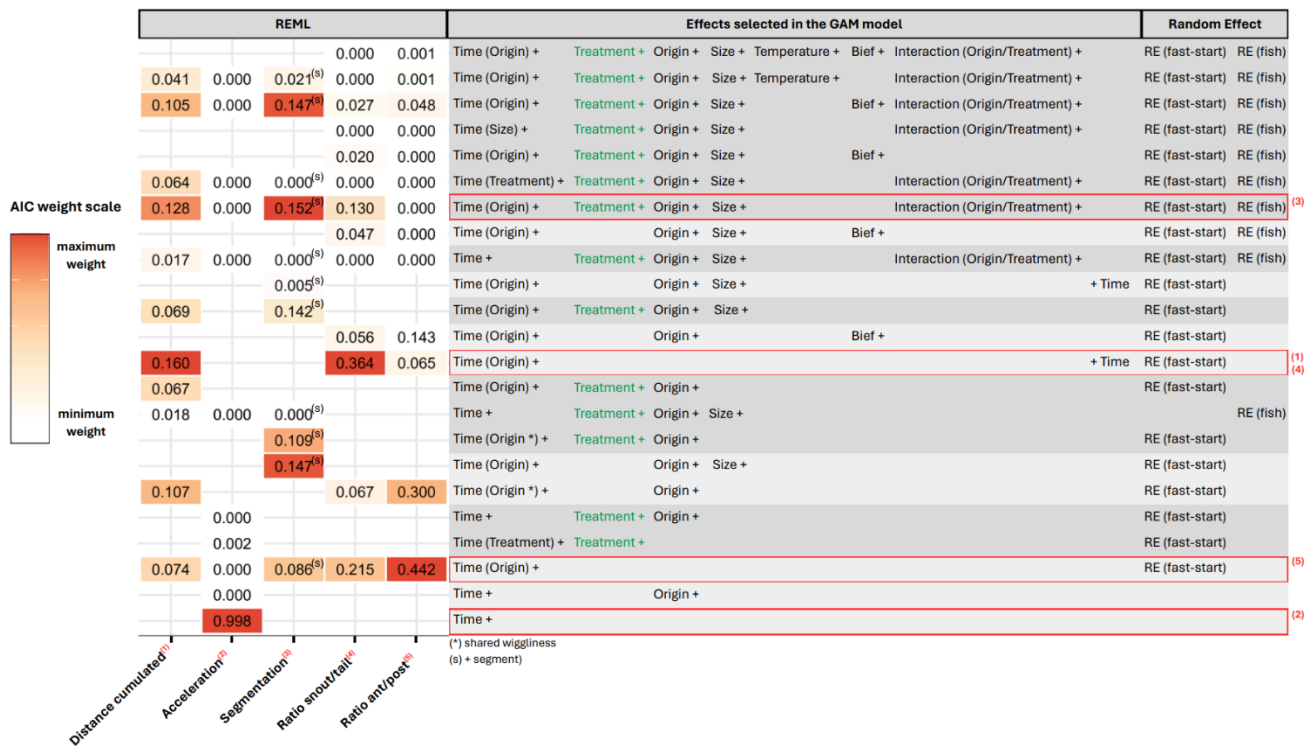


Figure 9: Comparative table of AIC weights of models fitted for the five time-related variables measured during fast-starts events using GA(M)M and REML method. Selected models are highlighted in red and treatment effect in green, (*) indicates temporal smooth sharing a common smoothing penalty, (s) indicates that models fitted for 'segmentation' integrated the effect of the segment number. All models were fitted with at least a smoothed time effect.

Along with the predictions of the best fitted models, we modeled the differences in prediction between the significant terms to allow better understanding of the temporal dynamics. Estimated differences indicate a significant difference between the two trajectories when the IC95% does not include 0. The selected GAMs (except for the acceleration) indicated a non-linear smooth interaction between time and the genetic origin, allowing each group to have its own temporal trajectory and its unique smooth parameter. A random effect associated with fast-start identification was also retained. The random effect term included in the models accommodate the non-independence of fast-start measure, with the correct data structuration: fish > fast-start > frame, reducing temporal bias for repeated observations from the same fast-start. We sorted models using AIC and AIC weights. We considered models equivalent if $\Delta AIC < 2$, rather different if $2 < \Delta AIC < 5$, and really different if $\Delta AIC > 7$ (Burnham & Anderson, 2004). We always selected the lowest AIC and the most parsimonious model between two equivalent ones.

Predictions for the cumulated distance have been obtained from the GAM model with significant non-linear smooth interaction between time and genetic origin and the fast-start as a random effect ([Fig. 10](#)). The relationship between distance cumulated along the time consists of a common linear trend shared by all groups (edf=6.89) and genetic group-specific smooth deviations (edf=1.00 and 2.05). The model explained 93.0% of the total deviance, indicating high explanatory capacity. The estimated degrees of freedom suggested a good fit for non-linear smooth relation with time.

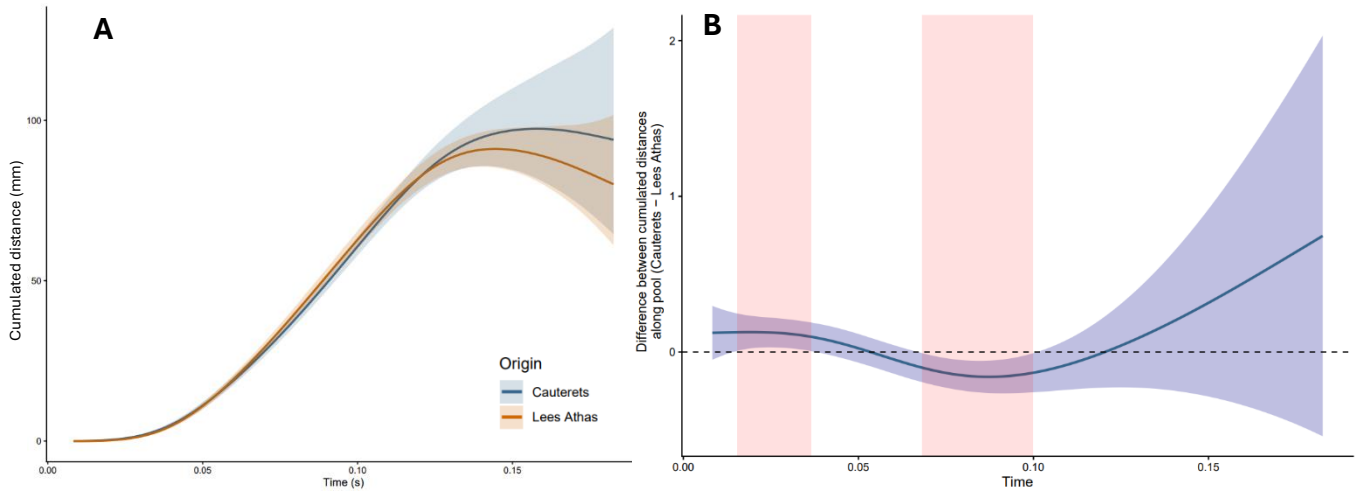


Figure 10: Illustration of the GAM selected for the cumulated distance in mm. **A)** Predictions of non-linear relations between the evolution of the cumulated distance achieved during fast start across time ($\Delta\text{frame} = 0.0083\text{s}$) for the two genetic origins. IC95% is visible for each pool. **B)** Estimated difference between the temporal trajectories of the cumulated distance achieved along the pool (Cauterets - Lees Athas). Significant temporal differences are highlighted in red, where the IC95% difference does not include 0.

The graph on the left ([Fig.10A](#)) shows the prediction of the temporal dynamics of the cumulated distance achieved. We can observe that the majority of differences also tend to appear late in the fast start, in the second phase. However, the second panel ([Fig.10B](#)) indicates that the two trajectories are significantly different between 0.015s to 0.035s and 0.070s to 0.10s where the IC95% does not meet the zero axis. It does not suggest significant divergence after 0.10s. In the first time-interval (0.015-0.035s) the Cauterets pool is likely to cover more distance, and the opposite goes for the second interval (0.070-0.10s). The table of criteria selection to choose the best model is shown in ([Appendix 13](#)).

About the modelling of the acceleration across time, the best model fitted only kept the non-linear relation with time as significant (edf = 6.27) without any other effects. The model explained 15.5%

of the total deviance, indicating low explanatory power. The table of criteria selection is in ([Appendix 14](#)).

Predictions for the cumulated difference of angle per segment ('segmentation') have been obtained from the GAM model with significant non-linear smooth interaction between time and genetic origin, the size, the segment identification as fixed effects and the fast-start as a random effect. The model explained 83.1% of the total deviance, indicating high explanatory capacity, but with 3 explicative terms and one more as a random effect. The estimated degrees of freedom suggested a good fit for a non-linear smooth relation with the interaction between time and genetic origin (edf=7.45 and 8.50). Here we only represented predictions for one segment ([Fig. 11](#)).

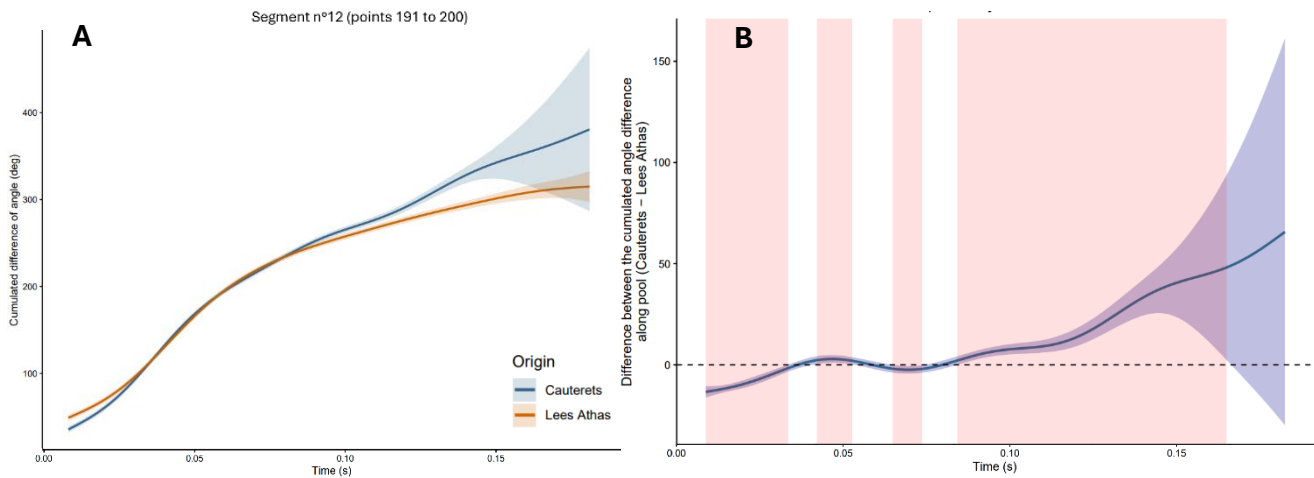


Figure 11: Illustration of the GAM selected for 'segmentation'. **A)** Predictions of non-linear relations between the evolution of the cumulated difference of angle (deg) for the last unique segment corresponding to the extreme tail part during fast start across time ($\Delta\text{frame} = 0.0083\text{s}$) for the two genetic origins. IC95% is visible for each pool. **B)** Estimated difference between the temporal trajectories of the same cumulated angle along the pool (Cauterets - Lees Athas). Significant temporal differences are highlighted in red, where the IC95% difference does not include 0.

The graph on the left ([Fig.11A](#)) shows the prediction of the temporal dynamics of the cumulated angle difference for the last segment n°12, corresponding to the extreme tail part shows the maximum cumulated difference. We can observe that the majority of differences seem to appear late in the fast start, in the second phase (refer to second-bend). The second panel ([Fig.11B](#)) that shows estimated differences helps us to better visualize the significant time-related differences between the two trajectories. We can detect differences in early stages (10-35ms, 45-55ms, 70-75ms), especially at the start, but after 3ms the general curves remain close suggesting limited evidence for important divergence. Indeed, the majority of the difference in terms of angle variation appears after 100ms. For both pools, the 12th segment shows high-angle differences in the early stages: 190° variation in 50ms. Cauterets pool differs after the 80ms, showing higher

variability in curvature of segments. The table of criteria selection to choose the best model is shown in ([Appendix 15](#)) and the predictions for all of the segments are shown in ([Appendix 16](#)).

Angular ratios calculated between the snout and the tail, and for the anterior and posterior parts were both fitted with relatively similar GAM. Both ratios captured rather similar information and had a close distribution ([Fig. 12](#))

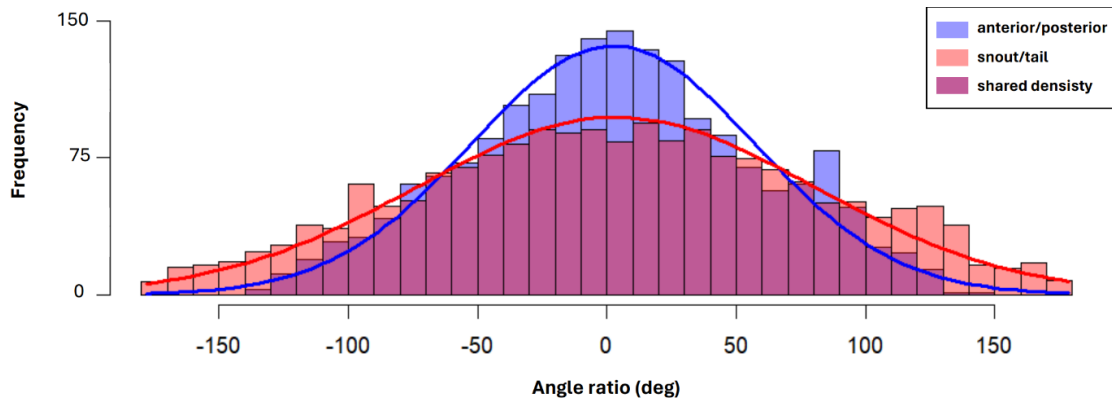


Figure 12: Comparison histogram of the distribution of angle ratio values calculated at each instant for both anterior/posterior and snout/tail. Normal curves have been added for visualization blue for anterior/posterior and red for snout/tail. Signals were highly correlated as expected (Pearson correlation; $t = 154.21$; $df = 246$; $p < 2.2e-16$; $cor = 0.95$).

Predictions for the snout-tail ratio have been obtained from the GAM model with significant non-linear smooth interaction between time and genetic origin and the fast-start as a random effect ([Fig. 13](#)). The relationship between distance cumulated along the time consists of a common linear trend shared by all groups ($edf=7.54$) and genetic group-specific smooth deviations ($edf=0.31$ and 7.60). The model explained 42% of the total deviance, indicating low explanatory capacity. The estimated degrees of freedom suggested a good fit for non-linear smooth relation with the interaction time/pool ($edf=6.99$ and 8.69).

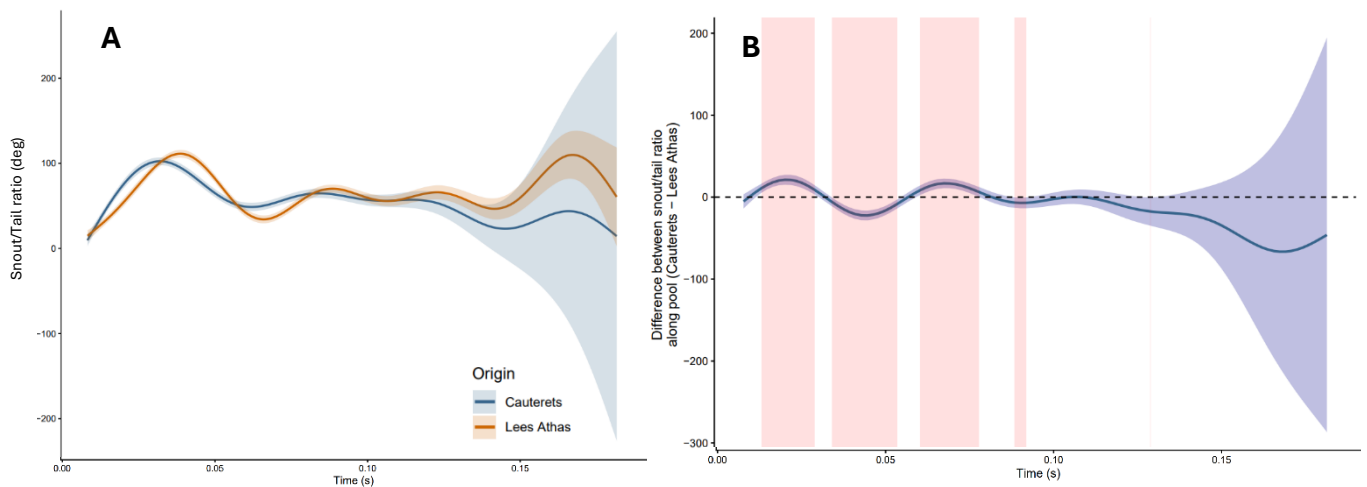


Figure 13: Illustration of the GAM selected for the angular ratio snout/tail. **A)** Predictions of non-linear relations between the evolution of the snout-tail ratio during fast start across time ($\Delta\text{frame} = 0.0083\text{s}$) for the two genetic origins. IC95% is visible for each pool. **B)** Estimated difference between the temporal trajectories of the snout-tail ratio along the pool (Cauterets - Lees Athas). Significant temporal differences are highlighted in red, where the IC95% difference does not include 0

The graph on the left ([Fig. 13A](#)) shows the prediction of the temporal dynamics of the ratio between the snout angle and the tail angle. This time, we can observe that the majority of differences tend to appear early in the fast start, in the first phase. The second panel ([Fig. 13B](#)) confirms that the two trajectories are significantly different between the intervals 15-35ms, 40-55ms, 80-85ms where the IC95% goes slightly over and under the zero axis. It does not suggest significant divergence after 100ms. On the first time-interval the Cauterets pool is likely to show more variation in terms of angle ratio, the opposite goes for the second interval and finally shows more variation for the third interval. Looking at the first graph, we can see that the signals are really similar, but Lees Athas appears to be late on the first peak and shows more wiggleness. The table of criteria selection to choose the best model is shown in ([Appendix 17](#)).

Predictions for the anterior-posterior ratio have been obtained from the GAM model with significant non-linear smooth interaction between time and genetic origin and the fast-start as a random effect ([Fig. 14](#)). The model explained 44% of the total deviance, indicating moderate explanatory capacity. The estimated degrees of freedom suggested a good fit for non-linear smooth relation with the interaction time/pool (edf=7.05 and 8.63). As for the ratio between the anterior and posterior part angles, both graphs ([Fig. 14](#)) appear very similar to those for the snout-tail ratio.

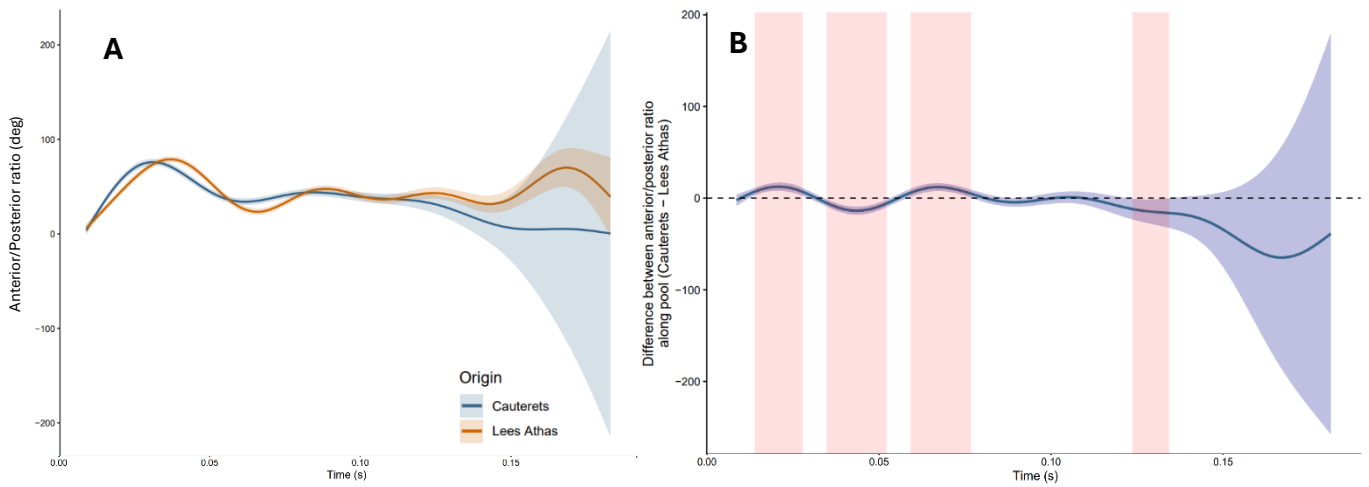


Figure 14: Illustrations of the GAM selected for angular ratio anterior/posterior. **A)** Predictions of non-linear relations between the evolution of the anterior-posterior ratio during fast start across time ($\Delta\text{frame} = 0.0083\text{s}$) for the two genetic origins. IC95% is visible for each pool. **B)** Estimated difference between the temporal trajectories of the anterior-posterior ratio along the pool (Cauterets-Lees Athas). Significant temporal differences are highlighted in red, where the IC95% difference does not include 0

The panel on the left ([Fig. 14A](#)) shows the prediction of the temporal dynamics of the ratio between the snout angle and the tail angle. Observations are very similar, on the first time-interval the Cauterets pool is likely to show more variation in terms of angle ratio, the opposite goes for the second interval and finally shows more variation for the third interval. We can observe that with anterior-posterior ratio, Lees Athas shows less wiggleness, and both IC95% are thinner than the predictions with the snout-tail ratio. The table of criteria selection to choose the best model is shown in ([Appendix 18](#)).

We tested for differences in efficiency during fast start according to treatment and genetic origin ([Fig. 15A](#)). Preliminary analyses revealed heterogeneity of variances between origins (Levene's test: $F_{1,190} = 7.30$, $p = 0.0075$), whereas treatment alone did not affect variance structure ($p = 0.92$). A generalized least squares (GLS) model allowing for different variances between origins was fitted using REML method with a principal effect of the interaction between treatment and genetic origin. Heteroscedasticity between origins was accounted for using a variance structure allowing different residual variances for each population. The model identified a significant interaction between treatment and origin (GLS; $F_{1,188} = 5.67$, $p = 0.018$), while neither treatment nor origin showed significant main effects, indicating that the final result differs between treatments depending on the genetic origin. Efficiency increased from upstream to aval in the Cauterets pool, whereas the opposite trend was observed in the Lees Athas pool ([Fig. 15B](#)). Although the

magnitude of these differences remained moderate, the crossing trajectories clearly illustrate the contrasting responses of the two populations to treatment.

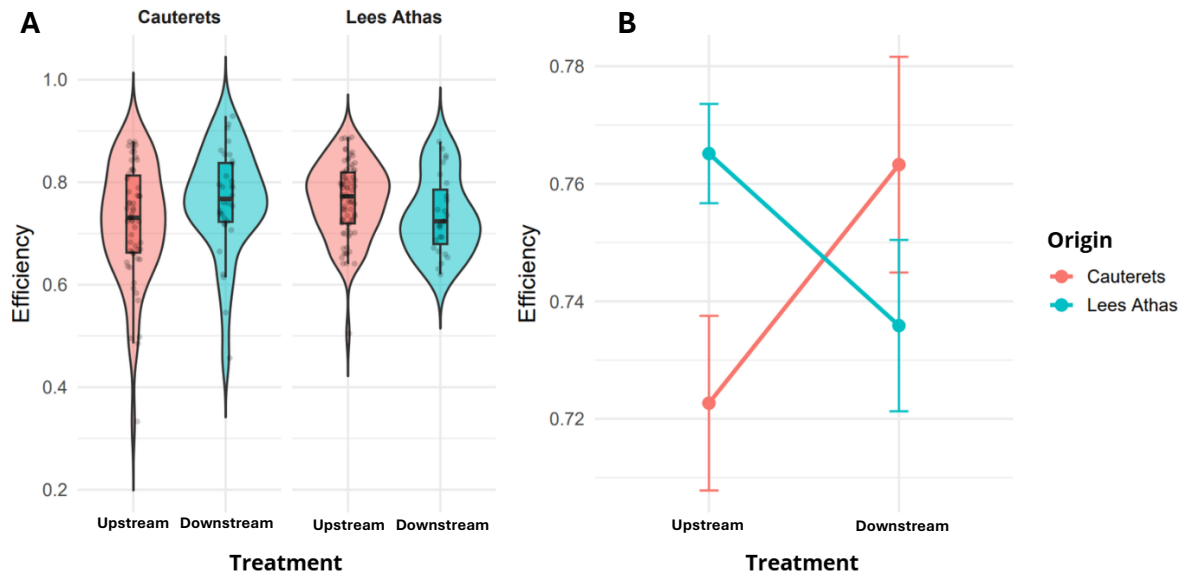


Figure 15: Illustration of the GLS model. **A)** Violin boxplots showing the distribution of efficiency values for each treatment within each genetic origin. **B)** Representation of the estimated values of efficiency and IC95% obtained from the GLS model.

4. Discussion

As largely known in the literature, water motion and flow regime have been shown to be a major pressure on the distribution of morphotypes and phenotypes of species based on their capacity to cope especially with intense flow (Langerhans and Reznick, 2009; Binning *et al.*, 2015). Inter-specific in particular studies are well documented, especially in coral reef fish assemblages and show high distinction of morphotype evolution and performances depending on wave exposure and water flow (Fulton *et al.*, 2005; Langerhans, 2008; Yan *et al.*, 2013; Binning and Roche, 2015). Intra-specific studies on the other hand, are limited possibly because of less direct morphological variations and similar morphotypes during experiments. Nonetheless, flow regime is still shown to be of critical importance during morphological development of fish (Imre *et al.*, 2002; Langerhans, 2008; Fischer-Rousseau *et al.*, 2010; Binning *et al.*, 2015). Salmonidae family in particular is commonly referenced as a biological model with good swimming capacities and also because water velocity is one of the most important environmental features characterising their habitat (Heggenes and Saltveit, 1990; Pakkasmaa and Piironen, 2000). It is known to present inter and intra-specific morphological differences due to phenotypic plasticity expression during juvenile development while coping

against different flow regimes and velocities (Pakkasmaa and Piironen, 2000; Domenici *et al.*, 2008; Jones *et al.*, 2020), as well as differences in swimming performances, physiology, fitness and dispersion that can be impacted by extreme flows (Jonsson and Jonsson, 2009; Binning *et al.*, 2015). Because morphology constraints performances (Koehl, 1996), functional traits and performances are highly dependent on morphological development (morpho-functionality). In this idea fish are not able to develop both features that optimize steady swimming and high unsteady performance (Langerhans, 2009; Ellerby and Gerry, 2011). Indeed, in fast-flow habitats, selection seems to favour more streamlined bodies with better steady swimming capacity to maintain position against the current, which often results in poorer fast-start performance compared to fish from slow-flow habitats (Fu *et al.*, 2013).

Overall, it is clear that the treatment (fluctuant vs. stable flow) did not affect intensely nor the steady/unsteady performances of individuals nor their behaviour in unaltered flow as our results did not show significant differences between treatment or clear distinctive variable that allowed treatment group discrimination (Fig. 9). Our results showed that the treatment individuals faced: more intense flow regime, up to 4 to 5 times the normal speed, 2 to 3 times a week during early life stage, was not inducing significant differences in terms of steady nor unsteady swimming performances. Pakkasmaa and Piironen (2000) showed significant results of morphological differences during the early developmental stage of the brown trout, having slightly slimmer shape in the fast water velocity with rather similar conditions of treatment (4 times the normal rate). One of the main differences between experiments was the experimental design where the environment consisted of long channels constructed from plastic ducts with no obstacles nor shelter, whereas in the experimental river, obstacles could be found under/behind gravel formations, grass and vegetation, and used as micro-shelters or microniches that could prevent the impact of treatment (Heggenes, 1988; Liao, 2007). Interesting results in accordance with our results come from another study from Flodmark *et al.* (2006) where fluctuating flow was tested on behaviour and growth with brown trout juveniles and showed no distinct differences despite daily important variation in flow (3 to 13 L.s⁻¹) and important reduction in current speed (0 to 0.87 m/s²). Their experimental design consisted of circular concrete tanks hosting 30 individuals each. The

differences in environmental conditions (artificial, semi-natural) could demonstrate that having more structurally-complex habitats with shelters, microniches can lower the impact of high flow regimes on juvenile stages. This idea claims for the conservation of biocomplexity.

Random forest analysis had a high classification error for the treatment suggesting low reliability of the classification returned with no strong descriptor to discriminate between the two groups. Results were slightly better in the analysis of the genetic origin classification with a lower error, and global results highly suggested an important interaction between the two condition treatments and genetic origin.

On the other hand, the interaction between time and genetic origin was almost always significant in our results of fast-start kinematics analysis, showing different temporal dynamics. In terms of distance achieved, it could mean that fish achieve different distances in each frame (lag of 8ms) depending on origin. Because fast-start are mainly triggered within prey-predator interaction where trout attend an escape of the attack, differences, even in very short distances or speed, can decide whether or not the fish gets captured and eaten ([Walker et al., 2005](#); [Nadler et al., 2021](#)). [Walker et al. \(2005\)](#) showed that a difference of one single standard deviation during fast-start performances could enhance the probability of surviving a strike 2 to 3 times on average. This difference between groups could suggest that the genetic origin has more importance than the treatment individuals faced during juvenile development. The effect of treatment was not important enough despite the genetic origin, suggesting that the majority of performance variation at the species scale is highly dependant on origin. Wild individuals were obtained from genitors captured the year before, while reared individuals were selected for more than 30 years ([Masso et al., 2018](#)), causing a potential domestication syndrome ([Bouchard and Abdallah, 2025](#)). This syndrome is known to cause important differences in fish behaviour and development, with less endurance ([Brown et al., 2013](#)), weaker performances and escape success ([Walker et al., 2005](#); [Vignon and Aymes, 2020](#)). In our results, genetic lineage is different in geographic localisation (Atlantic vs Mediterranean, [Fig. 1A](#)) and consists of reared individuals (Lees-Athas) and wild-origin ones (Cauterets). Statistical modelling showed consistent differences in fast-start performances depending on this origin, directly linked to escape success

and survival rate. The wild-origin batch was predicted to present more bending-variation in the early stages and more distance covered, suggesting higher chances of escape in case of predator strikes ([Walker et al., 2005](#)). In terms of acceleration, no discrimination was determined between treatment conditions or genetic origin, possibly suggesting no important differences in power/burst. Acceleration does not necessarily depend on neuromotor capacity, which is often measured with latency during stimulus responses ([Nadler et al., 2021](#)). Indeed, wild-origin individuals showed higher agility (bending capacity) that could favour selection in fast-starts events with the same linear performance capacity (speed, acceleration) with potentially increased unpredictable trajectories ([Domenici and Blake, 1997](#); [Langerhans, 2009](#)).

Also, there was an effect of treatment on the efficiency during fast-start, but inconsistent results of the genetic origin allowed no clear conclusion on the effect of treatment alone. Reared-individuals demonstrated better efficiency when they faced fluctuant regime conditions (upstream) but weaker capacities with a constant flow regime (aval). Differences in efficiency were about 4% when facing upstream conditions, indicating a position difference of approximately 0.4-0.5 cm (mean distance achieved being around 10-12 cm). As said earlier, even the slightest difference in fast-start performance could suggest a high difference in escape rate, difference in escape outcome, and hence in survival. In this idea, reared individuals showing better efficiency while coping against fluctuating flow regime could show better escape rate and fitness than wild-origin individuals. With the domestic syndrome, reared individuals usually demonstrate better competitiveness with increased aggression and dominance in a stable environment ([Bouchard and Abdallah, 2025](#)), and without the presence of predators, they could severely impact the development of other wild-origin individuals ([Bohlin et al., 2002](#)). Regarding our results of efficiency analysis, reared individuals show better capacity in stressful conditions and the opposite in the more stable environment, suggesting that the competitiveness induces an accelerated growth and early maturity at the expense of energy conservation and predator avoidance ([Bouchard and Abdallah, 2025](#)) during juvenile development is not the major driver of efficiency during fast-start, nor agility performances.

The comparison of the functional trade-off between steady (cruising) and unsteady (burst/escape) swimming was difficult to assess in this study as the number of individuals measured was only of $N=23$. Due to the nature of the fish's locomotor system, where the same structures power both modes (Blake, 2004; Langerhans, 2009), individuals that showed better fast-start performances, possibly wild-origin individuals regarding bending and cumulated distance differences, should demonstrate lower performances in steady performances (cruising against the flow). Supplementary analysis conducted with M. Descat returned that i) using PLS regression, the correlation between the latent variables extracted from the two steady/unsteady blocks was of 33% but still remained insignificant (PLS; $r_{\text{PLS}}=0.33$; $Z=(-0.29)$; $p=0.61$). This correlation is of high importance, but due to the low number of observations, it is to be further analysed to conclude on possible trade-off variation. ii) There was no significant relationship between morphological variability and the treatment conditions (Appendix. 19), as expected with important relationship between morphology and functional performance traits (Koehl, 1996). iii) using PLS regression, the correlation between variables from morphology analysis and swimming analysis (behaviour, fast-start, steady swimming respectively) were moderate (PLS; $r_{\text{PLS}}=0.17$; $r_{\text{PLS}}=0.37^{**}$; $r_{\text{PLS}}=0.53^{**}$) and returning that the first axis were described mainly by the *tail-beat frequency* during steady swimming, and the *segmentation* during fast-start indicating potential in more precise analysis of kinematics.

In the idea of better characterising individual behaviour, we try to determine possible behavioural syndromes. It is defined as the correlation between certain behavioural traits and possible typical profiles (Höjesjö et al., 2011). In previous work, Kern et al. (2016) found an interesting link between typical behaviour (shy/bold) and phenotype in fish suggesting that the behavioural syndrome integrated a broad suite of behaviour and functional traits. It was shown that pleiotropy occurred with a set of genes driving both a bold personality and directly enhancing the muscle power and the fast-start performance (Kern et al., 2016) In addition, as work is still in progress, we try to use Fast Fourier Transform (FFT) to detect movement periodicity and main swimming modes (oscillation frequency) during the 5 minutes of unsteady swimming. Differences in mode frequency could inform on stereotypical behaviour or be linked with other metrics. Main swimming modes of individuals inform us on their movements across the whole arena, going back and forth,

at which frequency, and about tail beat tendencies. Also, in addition to the time spent in each zone, we are trying to determine the number of entries/exits in each zone across time to differentiate fishes with the same frequency in the same zones but differ in terms of transition in between zones that could bring more information on the use of space by each individual (Flodmark *et al.*, 2006). The objective of detecting such details is to better characterise each individual's behaviour and possibly stereotypical pattern across all individuals that could be related to the treatment they faced.

Our results do not show an important effect of treatment alone, but the genetic lineage seems to have importance and is often cited in literature comparing reared vs. wild individuals. This is known to have important impact on conservation measures, where the main method employed to manage/preserve wild populations nowadays consists of restocking habitats by releasing eggs, juveniles, or adult individuals, often from pisciculture, directly into habitats of low density (intentional introduction) (Bohlin *et al.*, 2002; Bouchard and Abdallah, 2025). Although intended to boost declining populations, this practice often leads to serious genetic consequences, most notably the introgression of alleles from domesticated or non-native strains (Bouchard and Abdallah, 2025). Such genetic issues could compromise local adaptations, diminish the biological fitness of wild populations and result in an overall loss of genetic diversity. Our results tend to be in accordance with this belief that reared individuals demonstrate lower performances during fast-start actuation (Walker *et al.*, 2005; Brown *et al.*, 2013), which could in the long-term become harmful for wild populations.

In conclusion, regarding the sizes of the effect and the importance of genetic/treatment interaction those conclusions are to be moderated. Results indicate high developmental robustness during the ontogeny of fishes coping with highly fluctuant flow. This robustness is maintained independently of the genetic origin when analysing morphology, steady swimming, and unsteady behaviour, and appears with low differences when assessing fast-start performances. As explained earlier, different results can depend highly of the experimental conditions. Environmental conditions play a crucial role in providing shelters and refuges during floods where individuals can hide and protect themselves against intense events. Floods' effect

can be partially countered with complex habitats, but other events such as droughts are likely to offer such refuges with important decrease in water level, fully exposing individuals to stressful conditions. Therefore, assessing the impact of hydrological natural events intensified by the global change on fish morphology, behaviour and functional traits, needs to integrate a correct description of habitats, different hydrological events with potential higher intensity especially to detect fine variations at the micro-evolutive scale. Although having low differences in performances, using reared individuals from pisciculture to repopulate habitats through intentional introduction still causes genetic issues in the long term and can not be supported by our results.

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6. Software and macro references

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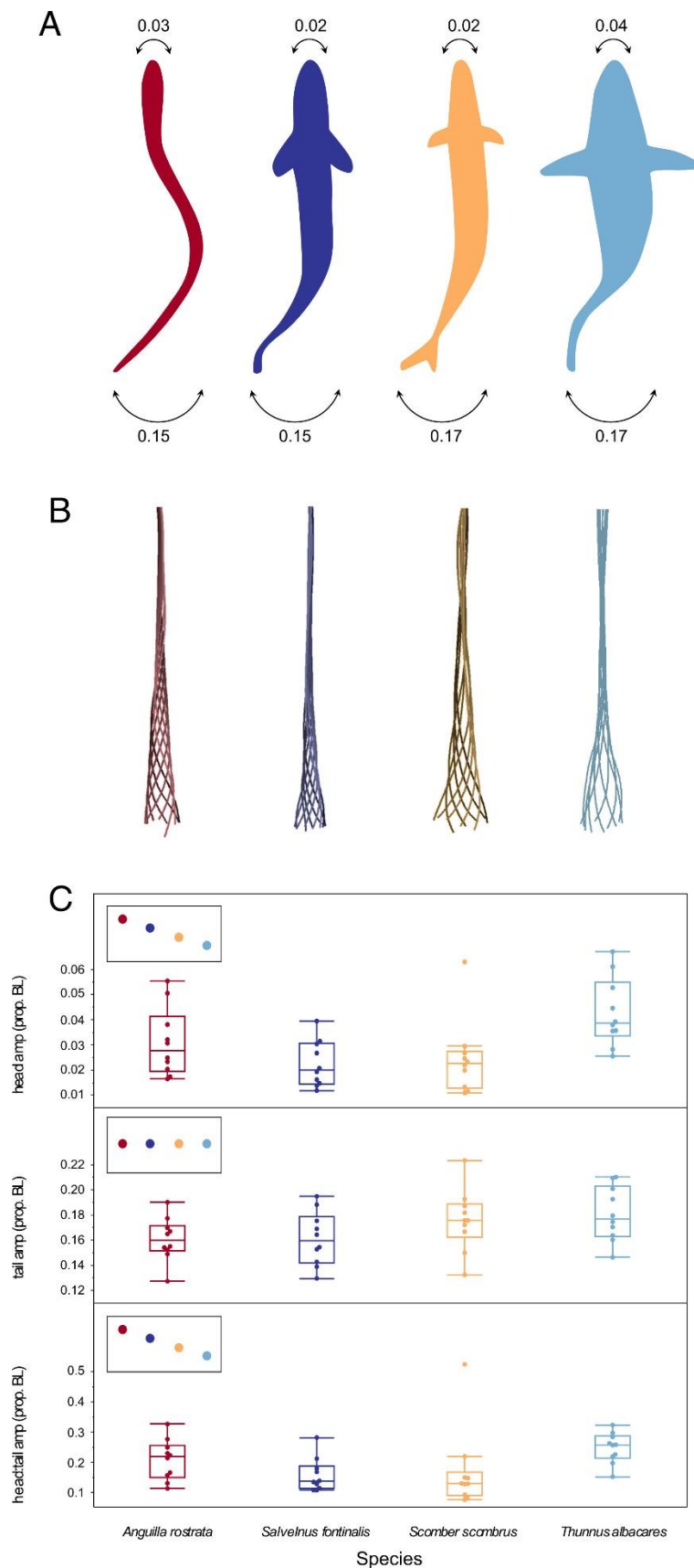
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7. Appendices

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Appendix 1: Undulatory swimming modes in fishes.



Rethinking swimming modes in fishes.

(A) New view based on high-speed video analysis of steady swimming kinematics to illustrate the considerable similarity among the four classic fish swimming modes. Head and tail amplitude variables do not distinguish the four classic categories of BCF swimmers: eel (*Anguilla rostrata*), trout (*Salvelinus fontinalis*), mackerel (*Scomber scombrus*), and tuna (*Thunnus albacares*). **(B)**

Representative midlines (average of $n = 10$ each) of complete tail beat cycles of eel, trout, mackerel, and tuna during steady swimming. **(C)** Distribution of head, tail, and head:tail amplitude reported as a proportion of BL in four representative species ($n = 10$ clips per species). The boxes show the median as well as the 25th and 75th, while whiskers show the 10th and 90th. (Top Left Insets) Expected pattern based on the canonical classification of swimming modes. Original figure from Di Santo et al., 2021.

Appendix 2: Experiences description and protocol conducted by M. Descat.

i) Experimental design: When 80% resorption of the vesicle was achieved, individuals were inserted in open small container with gravel (acclimatation for one month) in the semi-natural flume (INRAE, 1980) where trout developed with two different treatments for three months: upstream group with high variations of flow regime (fluctuant environment) and downstream also called aval with a lower consistent flow (stable environment). Variation in flow occurred 2 to 3 times a week and was 4 to 5 times higher in terms of intensity (Fig. 16). The trout developed in semi-natural conditions with water sourced from the river *Lapitxuri*, a natural affluent of the *Nivelle* (located close to Dantxarinea - Navarre, French-Spanish border). As trout developed, 214 individuals were collected after the three months to test swimming patterns, the ~200 others remaining in the semi-natural flume.

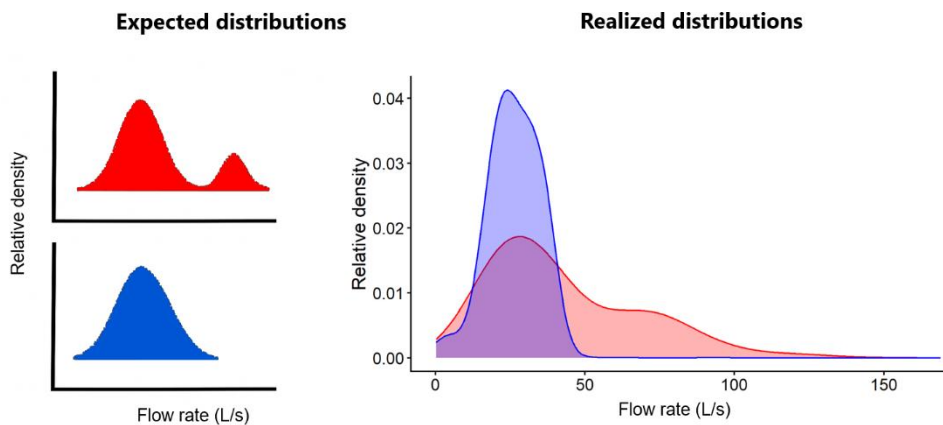


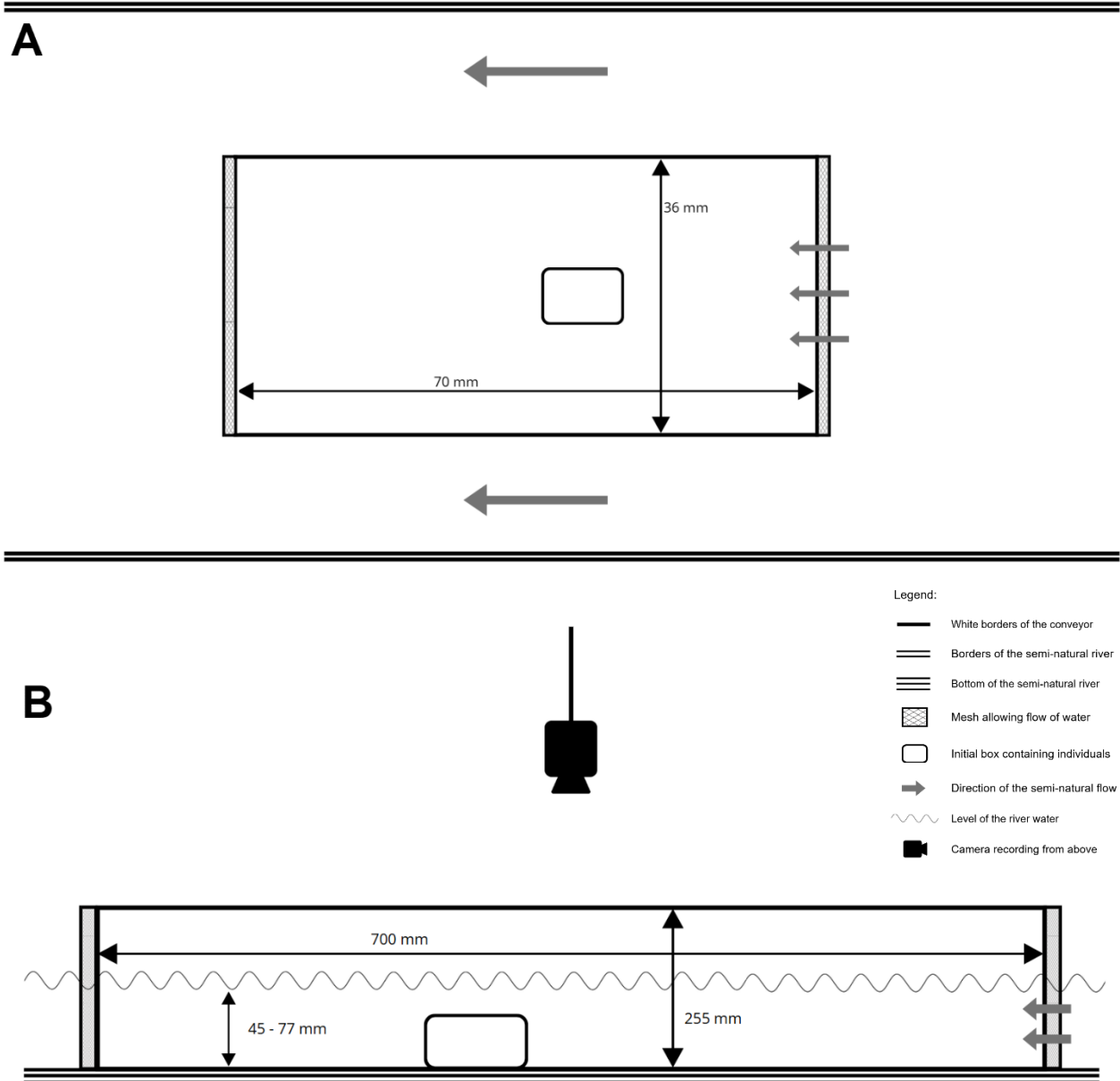
Figure 16: Density of the flow regime distribution for each treatment. From M. Descat.

ii) Steady swimming test design: Plastic conveyors were placed in the flume, closed by mesh at the front and the back, allowing water passage (Appendix 3). One by one, the 214 individuals were placed inside the conveyor and filmed for 20 minutes. The device allowed steady swimming patterns to be filmed.

iii) Unsteady swimming test design: Octagonal white tanks 40 cm wide (Appendix 5) where the same 214 fish were placed individually and filmed during 10 minutes. The first 5 minutes, fishes and water remained undisturbed to allow unsteady swimming patterns to be filmed. The next 10 minutes consisted of a response to external stimulus to induce fast-start. Stimulus was observers tapping against the wall to produce sound and wave disturbance (Appendix 5). About 3 to 5 fast-starts were recorded per fish. The first frame where movement is initiated and visible is therefore attributed as the second frame of movement, with maximum precision of $1/120 = 8$ ms.

iv) Videos Set-up: Videos were recorded in 40 fps, 1640*922 resolution for the first experience (Appendix 3) and in 120 fps, 958*720 resolution for the second (Appendix 5) to allow kinematics analysis of fast-start that lasts about 0.2 seconds. Cameras (Raspberry PiScope designed by INRAE technician) were recording from above in both experiences (Appendices 3, 5). The recording started right before individuals were placed inside the experimental set-ups.

Appendix 3: Experimental setup for **steady swimming** analysis.



Experimental setup for the **steady swimming analysis** (with flow). **A)** Visualisation from above with dimensions of the plastic conveyor. **B)** Visualisation from the side with camera setup. The box containing individuals was placed in the centre and let untouched for 10 minutes. After, Individuals were placed in the conveyor and the box was removed if the individual had not already escaped by itself. The flow of the water was controlled to stay at a relative level (between 45 and 77 mm), allowing similar conditions for each individual. The camera was recording from above at 120 fps for 20 minutes in total per individual. The conveyor was installed inside the semi-natural flume to reproduce an unaltered flow with semi-natural conditions. Created with [Canva](#).

Appendix 4: custom macro “Segmented Action Tool – Multilines (cm)” for semi-automated digitalization of the body midline of fishes. Developed by Matthias Vignon for ImageJ. Replace ‘ $N = 16$ ’ by ‘ $N = 200$ ’ for method comparison.

```
macro "Segmented Action Tool - Multilines (cm)" {

    // Nombre de points fixe
    N = 16;

    // Crée la table uniquement si la fenêtre Results n'existe pas encore
    if (isOpen("Results") == 0) {
        run("Clear Results"); // créer table vide
    }

    getSelectionCoordinates(xpoints, ypoints);
    getPixelSize(unit, pixelWidth, pixelHeight);
    // pixelWidth et pixelHeight = taille physique d'un pixel (en cm si échelle définie)

    // Calculer la longueur totale de la ligne en cm
    l = 0;
    for (i = 1; i < xpoints.length; i++) {
        dx = (xpoints[i] - xpoints[i-1]) * pixelWidth;
        dy = (ypoints[i] - ypoints[i-1]) * pixelHeight;
        length = sqrt(dx*dx + dy*dy); // en cm
        l += length;
    }

    // Intervalle en cm
    ll_cm = l / (N - 1);

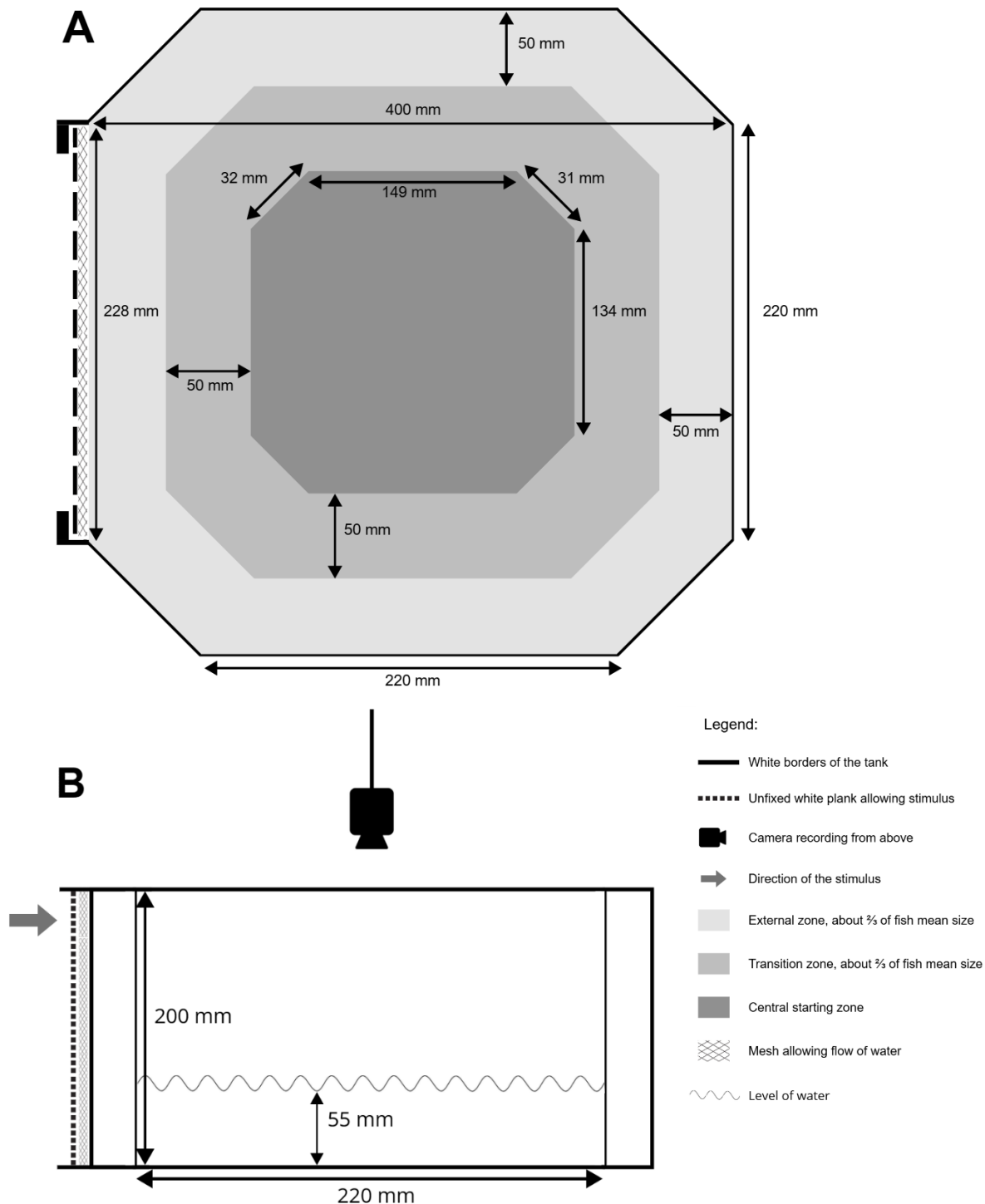
    // Conversion en pixels pour Interpolate
    // ⚠ suppose pixels carrés (pixelWidth ≈ pixelHeight)
    ll_px = ll_cm / pixelWidth;

    run("Interpolate", "interval=" + ll_px);
    getSelectionCoordinates(xpoints, ypoints);
    n = nResults; // nombre de lignes déjà présentes dans Results

    for (i = 0; i < xpoints.length; i++) {
        // Conversion des coordonnées pixels → cm
        Xcm = xpoints[i] * pixelWidth;
        Ycm = ypoints[i] * pixelHeight;

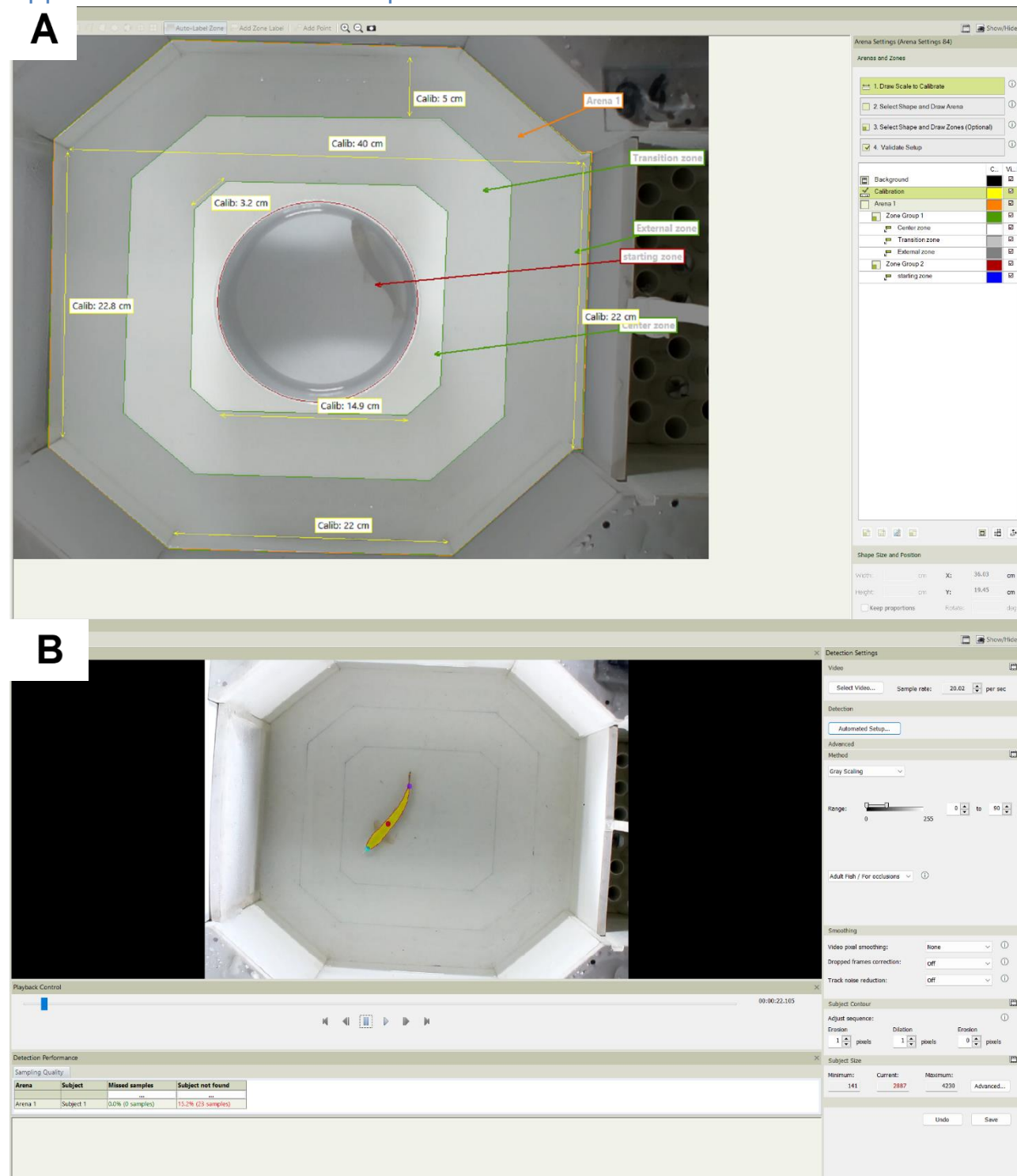
        // Ajouter les coordonnées en cm dans Results
        setResult("X (cm)", n + i, Xcm);
        setResult("Y (cm)", n + i, Ycm);
    }
    updateResults();
}
```

Appendix 5: Experimental setup for the **unsteady swimming** analysis.



Experimental setup for the **unsteady swimming** analysis (without flow). **A)** Visualisation from the top of the zone mapping for behaviour and thigmotaxis analysis. **B)** Visualisation from the side with wall heights, with the camera setup. The camera recorded from above, allowing full-scene tracking, at 120 fps. The stimulus to induce fast-start response consisted of one experimenter tapping the unfixed plank to produce wave and sound through the tank. Created with [Canva](#).

Appendix 6: Screenshot example of one calibration realized on Ethovision 18.



Examples of calibration using [EthoVision](#) for future behaviour analysis with zonation. In total 184 sequences of 5 minutes have been calibrated for future analysis. **A)** Arena settings with zonation and distance calibration. **B)** Detection settings with the grey scaling method for 3 point detections. All parameters to calibrate the videos correctly are available as supplementary data on request.

Appendix 7: Protocol and modifications for CurveMapper4

The current script was downloaded from: <<https://zenodo.org/records/4623882>>

Direct download link <[CurveMapper6.m](#)>

The file 'CurveMapper6.m' contain the function 'CurveMapper4'

We used the protocol as described in the explanation document,

Direct download link <[CurveMapper5.doc](#)>

However, to adapt the script to our data we made slight modifications to redirect output files as explained in the following description:

```
#####  
# File modification 'CurveMapper4.m'  
#####
```

line 387, in the “else” part, replace:

```
else  
    s.FileName= FileName;  
    s.PathName= PathName;  
    s.OutputFileName= [s.FileName '_CURVES.xls'];  
    s.OutputPathName= s.PathName;  
    s.MovObj= VideoReader([PathName FileName]);
```

by:

```
else  
    s.FileName= FileName;  
    s.PathName= PathName;  
    %MODIFICATION : next line is added to define the name wihtout '.avi'  
    [~,fileNameNoExt, ~] = fileparts(FileName);  
    %MODIFICATION : Original line -> s.OutputFileName= [s.FileName '_CURVES.xls'];  
    s.OutputFileName= [fileNameNoExt '.xls'];  
    s.OutputPathName = 'C:\ [your_path_here] \data_matlab\';  
    %MODIFICATION : Original line -> s.OutputPathName= s.PathName;  
    s.MovObj= VideoReader([PathName FileName]);
```

and replace [your_path_here] by the path to your root folder to save directly ouput files in 'data_matlab'

line 488, to avoid saving unnecessary (in our project) supplementary files that requires power and long-time processing, replace:

```
saveas(gcf,[s.OutputPathName s.OutputFileName '.jpg'],'jpg')
saveas(gcf,[s.OutputPathName s.OutputFileName '.emf'],'emf')
```

by:

```
%MODIFICATION : added percentage to avoid loading files not needed (jpg and emf)
%saveas(gcf,[s.OutputPathName s.OutputFileName '.jpg'],'jpg')
%saveas(gcf,[s.OutputPathName s.OutputFileName '.emf'],'emf')
```

line 533, to avoid saving unnecessary (in our project) supplementary files that requires power and long-time processing, replace:

```
saveas(gcf,[s.OutputPathName s.OutputFileName '.jpg'],'jpg')
saveas(gcf,[s.OutputPathName s.OutputFileName '.emf'],'emf')
```

by:

```
%MODIFICATION : added percentage to avoid loading files not needed (jpg and emf)
%saveas(gcf,[s.OutputPathName s.OutputFileName '.jpg'],'jpg')
%saveas(gcf,[s.OutputPathName s.OutputFileName '.emf'],'emf')
```

ligne 570, to avoid saving unnecessary (in our project) supplementary files that requires power and long-time processing, replace

```
saveas(gcf,[s.OutputPathName s.OutputFileName '.jpg'],'jpg')
saveas(gcf,[s.OutputPathName s.OutputFileName '.emf'],'emf')
```

by:

```
%MODIFICATION : added percentage to avoid loading files not needed (jpg and emf)
%saveas(gcf,[s.OutputPathName s.OutputFileName '.jpg'],'jpg')
%saveas(gcf,[s.OutputPathName s.OutputFileName '.emf'],'emf')
```

All modifications have been indexed with '%MODIFICATION' in the script to allow easier navigation through the custom script.

```
#####
# End of modification 'CurveMapper4.m'
#####
```

We resumed the protocol we use for our project using the guide document <[CurveMapper5.doc](#)>

Protocol (Available in supplementary data in 'Acquisition fast start'/README):

- Create a new folder (eg: 'acquisition_fast_start')
- Download CurveMapper4.m (<https://zenodo.org/records/4623882>)
- Modify the file 'CurveMapper4.m' (refer to the modifications part earlier)
- Use the following folder structure for easier organization and potential implementation in DLC:

A folder per day / a subfolder per hour-individual / one or more fast-start videos named 'ID_fs1' or 'ID_fs2' etc...

A folder 'data_matlab' with the output files from MATLAB that will be named 'ID_fs1' or 'ID_fs2' etc...

A folder 'labeled-data' / a subfolder per fast-start named 'ID_fs1' or 'ID_fs2' etc... containing all of the frames of the fast-start sequence.

The 'CurveMapper4.m' file

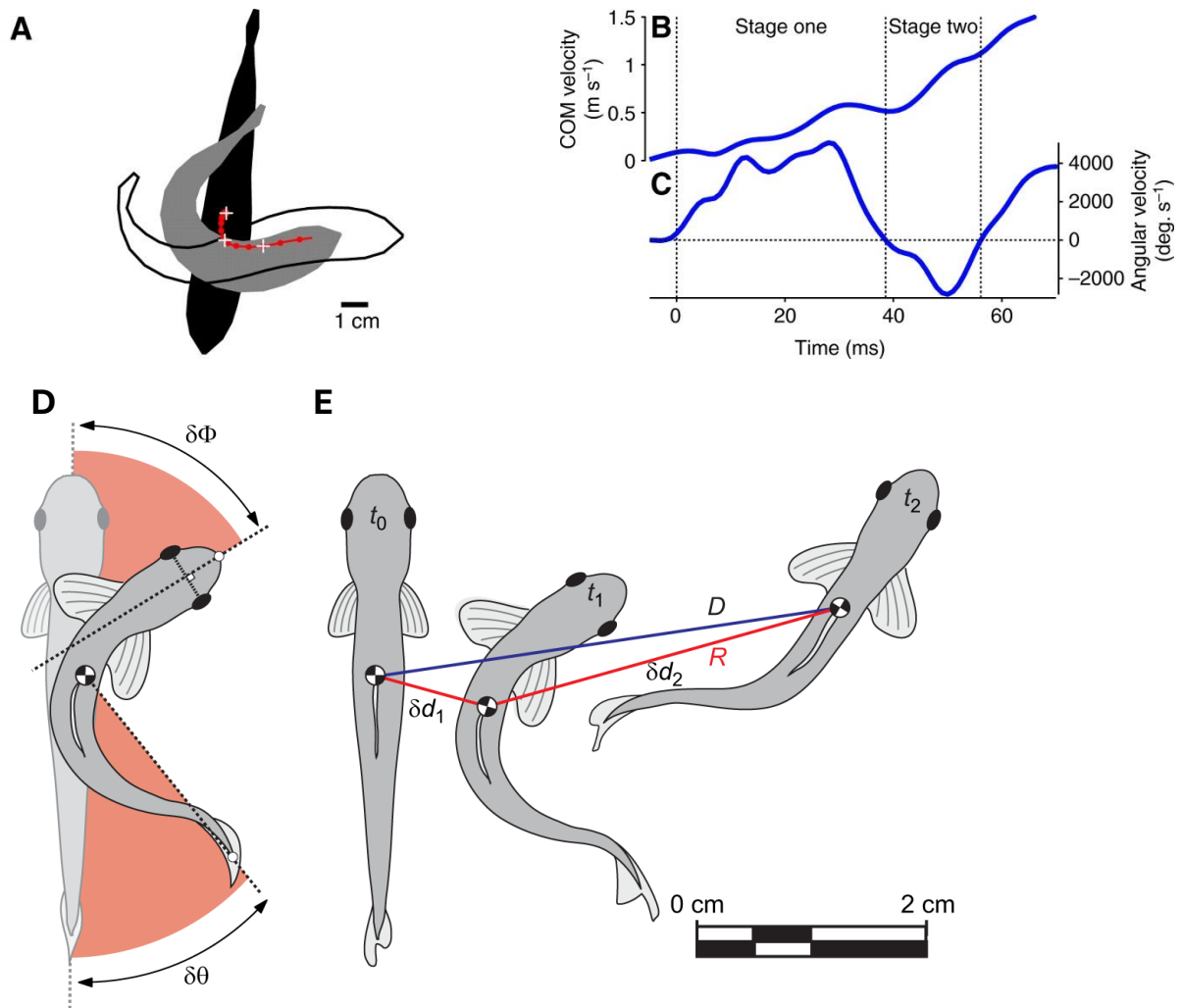
The 'CurveMapper5.doc' file for instructions

(optional) Custom file 'CurveMapper4Mod.m'

(optional) Custom file 'remove_png.py' pour supprimer les fichiers inutiles sorties par Matlab

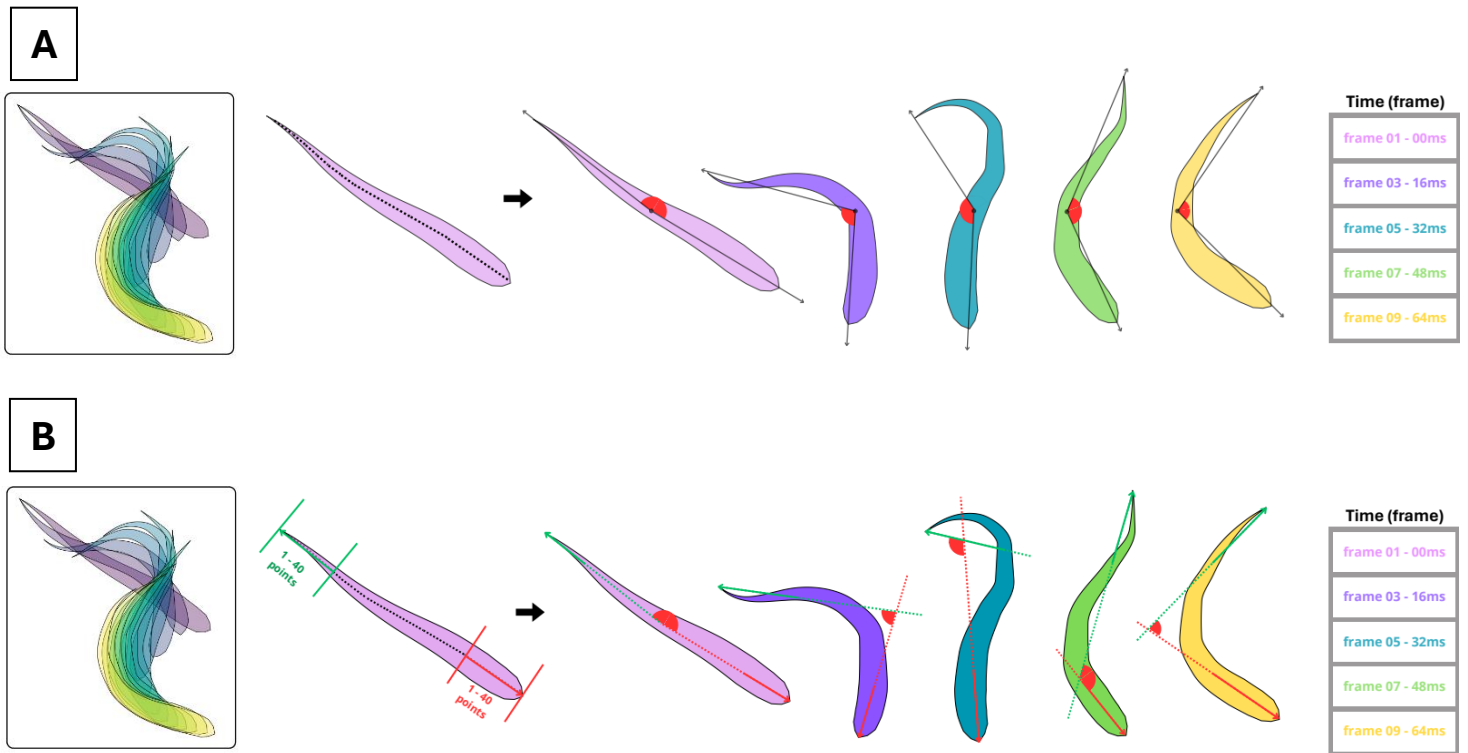
- Run MATLAB, wait for '>>' in console and call 'CurveMapper4'
- Select 'Multiple Frame'
- Select the video of interest
- Unselect 'Unit Conversion'
- Check that the output file is 'data_matlab'
- Check the correct naming of output file 'ID_fs1' or 'ID_fs2' etc...
- Start Frame : 1 ; Increment : 1 ; End Frame : check in 'labeled-data' the correct frame subfolder
- Run 'GO'
- 'Left mouse' to add point, 'Right mouse' to delete last point, 'Enter' to go to next frame
- Use '-' and '=' to zoom in/zoom out
- When finished, close image and open the next one

Appendix 8: Metrics measured during a fast start sequence.



Kinematics and metrics measured from a typical C-start. **A**) Silhouettes of the body in its initial position (black), at the end of stage one (gray) and at the end of stage two (open). The centre of mass (CM) is shown over time with a red line with dots every five milliseconds. White crosses mark the initial position and the positions of stage one and stage two. **B**) Velocity of the centre of mass over time. **C**) Angular velocity of the head over time. Divisions between stage one and two are shown with dotted lines. **D**) Fish outlines before (light grey) and after (dark grey) a bending response. The course of the head's angular changes ($\delta\Phi$) was measured as the angle of the anterior midlines (originating from the tip of the snout and intercepting the mid-distance point between the eyes) between two successive frames. The tail body curvature ($\delta\theta$) was measured as the angle of the posterior midlines (originating from the base of the tail to the centre of mass, CM) between two successive frames. **E**) Example of trout escaping (initial position on left). The CM (circled sights) are shown on each stage. The total distance travelled by the fish (R , sum of the distances covered by the CM between successive frames) is depicted using a red line, while the effective swim distance (D) is depicted in blue. The ratio between D and R is a measure of two-dimensional trajectory efficiency. **A**, **B**, **C** modified from Tytell and Lauder, 2008 ; **D**, **E** modified from Vignon and Aymes, 2020.

Appendix 9: Comparison of Angular ratio measurement for the fast-start kinematics

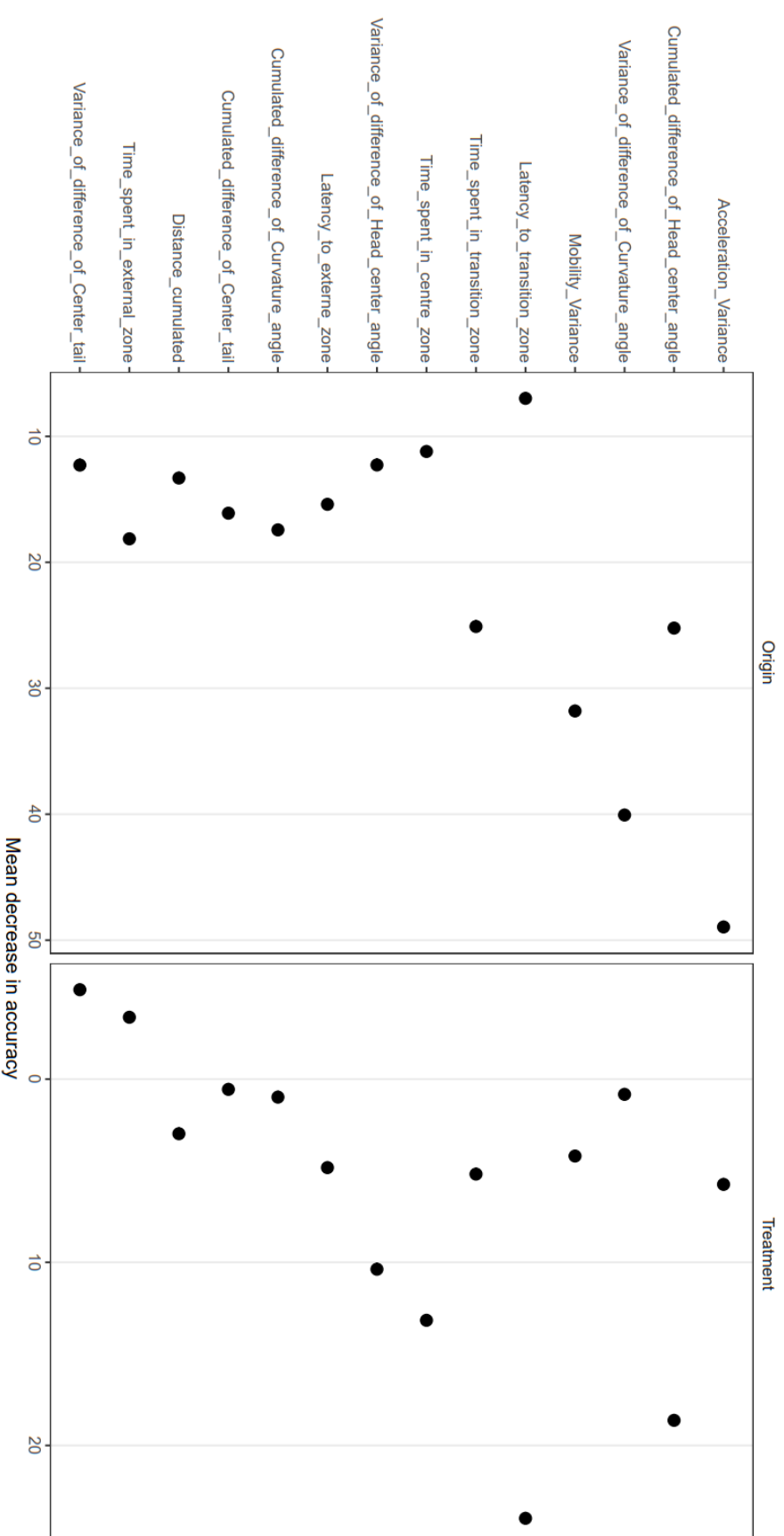


Fast start acutation is recreated on the left. The color corresponds to the time during the fast start event with a lag of 16ms. A) Visualisation of the anterior/posterior ratio calculated by the ratio formed with the 2 whole half of the fish. B) Visualisation of the snout/tail ratio calculated by the ratio formed with the 40 first points and the 40 last points digitalised

Appendix 10: Recapitulative table of metrics measured

		Metrics	Description	Performance/Trait
Unsteady swimming	EthoVision XT Behaviour analysis	Time spent in each zone	We calculate the time spent in each zone (s) per individual and make a percentage of it on the total duration (5 minutes)	The metrics for zonation indicates us on the use of space and about thigmotaxis
		Mobility	We calculate the percentage of mobility for the complete area per individual during a minimum time interval	Mobility can differentiate between two individuals that didn't move but were active in tail/head movement as a stress indicator
		Distance covered	We calculate the distance achieved (cm) at each minimum time interval	A direct measure of performance that informs us on exhaustion, durability, exploration
		Speed/Acceleration	We determine speed (cm/s) and acceleration (cm/s ²) for each minimum time interval	A direct measure of power and swimming performance linked to the individual capacities
		Head angle	We calculate the difference of angle (deg) achieved by the upper half of the individual body between two successive frames	A measure of the fish's head flexibility and activity during swim
		Tail angle	We calculate the difference of angle (deg) achieved by the lower half of the individual body between two successive frames	A measure of the fish's tail flexibility and activity during swim
		Curvature angle	We calculate the difference of curvature angle (deg) achieved between two successive frames	A measure of the fish's flexibility, bend and activity during swim
		Number of entries/exits	We determine the number of entries/exits between each zone	It helps differentiates between fishes that would spend the same time in the same zone but one goes in and out and the other stay still.
		Swimming modes (FFT)	While detecting a periodic signal (tail angle, curvature) we try to use the Fast Fourier Transformation to determine the main two modes of the signal	It could be useful to detect typical swimming mode (frequency) and patterns to better discriminate individuals
	MATLAB Fast-start analysis	Speed/Acceleration	We determine speed (cm/s) and acceleration (cm/s ²) for each minimum time interval	One of the most useful and direct measure of power and burst. Directly linked to the individual capacities, to overall and muscle, and the capacity to escape a rapid and fatal strike
		Distance covered	We calculate the distance achieved (cm) at each minimum time interval	A direct measure of performance that informs us on the capacity of the individual to escape a strike by achieving the maximum distance while dodging
		Efficiency	We calculate the difference between the total distance achieved from the start to end of fast start and the effective distance which represents the straight line between the first and last position	Efficiency informs us about the predictability of the fish fast-start (which path to use to go from the initial to last position)
		Segment Angle	We calculate the difference of angle (deg) achieved for each segment determined with the SGM, between two successive frames	A measure of the fish's flexibility and information on the importance of the different part of the midline during the fast-start actuation
		Curvature angle: anterior-posterior ratio	We calculate the difference of curvature angle (deg) achieved between two successive frames	A measure of the fish's flexibility and bend. The first maximum bending observed corresponds to the end of phase 1.
		Curvature angle: snout-tail ratio	We calculate the difference of curvature angle (deg) achieved between two successive frames only taking the first 20% and the last 20% of the body midline	A measure of the fish's flexibility, and bend, to see if taking extreme parts (40%) is enough to assess more or the same variability between each individual and fast-start.
Steady swimming	ImageJ	Tail-beat amplitude	We calculate the maximum amplitude achieved by the tail during steady swimming for 2 or 3 tail-beat sequences	The maximum amplitude is directly linked to the individual capacity to face different regime with a maximum trust

Appendix 11: Comparison table of the relative importance of each variable used in Random Forest classification for Treatment vs. Genetic Origin



	Fitted in the GAM model						Random Effect
	NVL			F1CVL			
	0.000	0.022	0.000	0.001	0.041	0.000	Time (Origin) + Treatment + Origin + Size + Temperature + Blot + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.000	0.000	0.000	0.001	0.000	Time (Origin) + Treatment + Origin + Size + Temperature + Blot + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.051	0.095	0.048	0.105	0.027	Time (Origin) + Treatment + Origin + Size + Blot + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.000	0.000	0.000	0.000	0.000	Time (Size) + Treatment + Origin + Size + Blot + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.037	0.061	0.000 ^(s)	0.000	0.064	0.020	Time (Origin) + Treatment + Origin + Size + Blot + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.000	0.000 ^(s)	0.000	0.000	0.000	Time (Treatment) + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.132	0.120	0.000	0.128	0.130	Time (Origin) + Treatment + Origin + Size + Blot + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.081	0.081	0.131 ^(s)	0.071	0.152 ^(s)	0.047	Time (Origin) + Treatment + Origin + Size + Blot + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.000	0.000 ^(s)	0.000	0.000	0.000	Time (Treatment) + Treatment + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.000	0.000 ^(s)	0.000	0.000	0.000	Time (Origin) + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.000	0.000 ^(s)	0.000	0.017	0.000 ^(s)	Time + Treatment + Origin + Size + Interaction (Origin/Treatment) + Time RE (fast start) RE (ish)
	0.000	0.013	0.000 ^(s)	0.000	0.005 ^(s)	0.000	Time (Origin) + Treatment + Origin + Size + Blot + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.066	0.126 ^(s)	0.069	0.142 ^(s)	0.056	Time (Origin) + Treatment + Origin + Size + Blot + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.084	0.237	0.078	0.143	0.160	0.364	Time (Origin) + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.019	0.027	0.065	0.067	0.000 ^(s)	Time + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.064	0.000 ^(s)	0.018	0.109 ^(s)	0.147 ^(s)	Time (Origin) + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.016	0.115 ^(s)	0.000 ^(s)	0.147 ^(s)	0.067	Time (Origin) + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.199	0.103	0.300	0.107	0.067	Time + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.000	0.000	0.000	0.000	0.000	Time (Treatment) + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.000	0.000	0.000	0.000	0.000	Time + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.002	0.289	0.069	0.442	0.074	0.086 ^(s)	Time (Origin) + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.074 ^(s)	0.339	0.000	0.086 ^(s)	0.215	Time (Origin) + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.000	0.000	0.000	0.000	0.000	0.000	Time + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)
	0.998	0.000	0.000 ^(s)	0.998	0.000	0.000	Time + Treatment + Origin + Size + Interaction (Origin/Treatment) + RE (fast start) RE (ish)

Appendix 13: Table of criteria model selection for cumulated distance

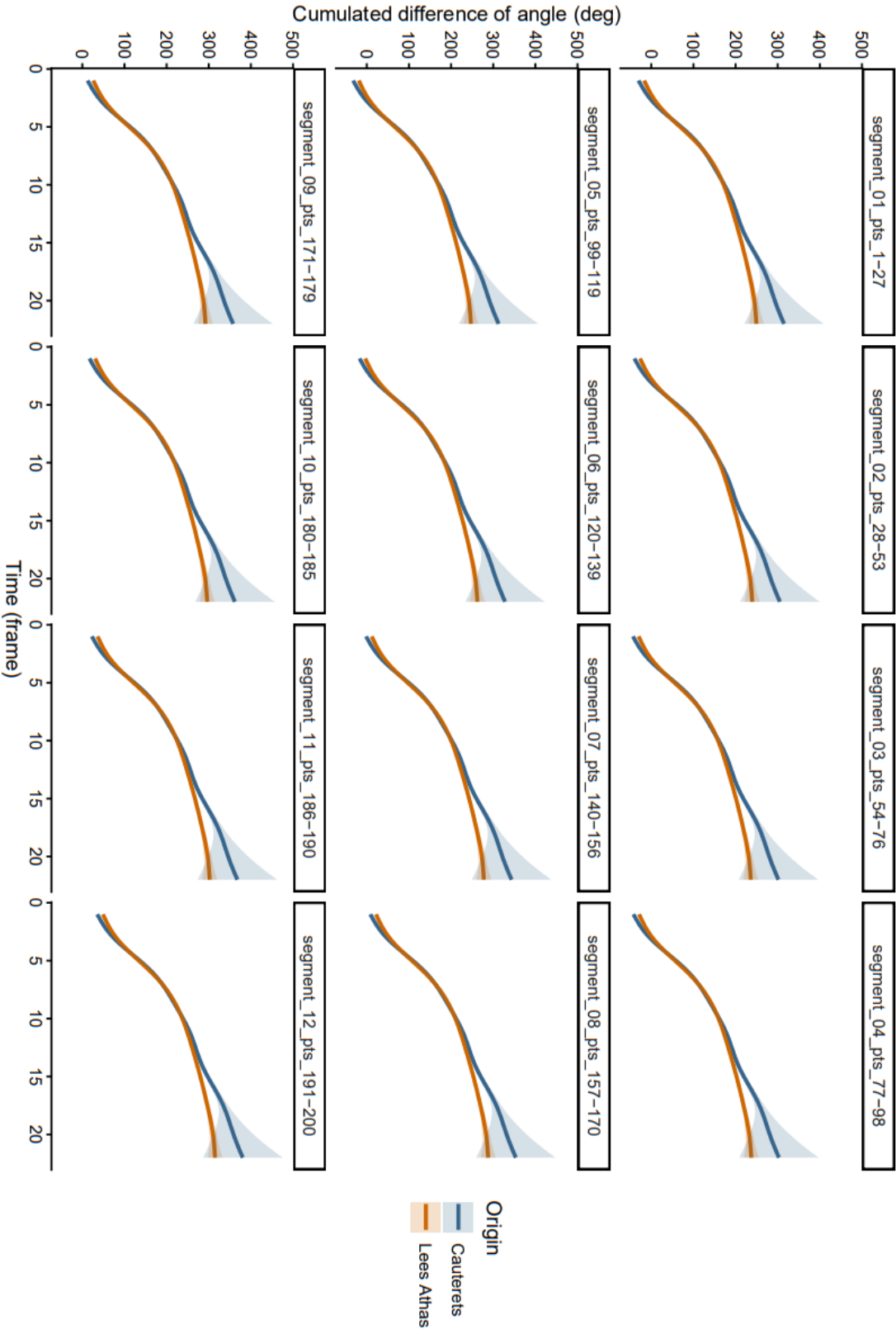
model_type	formula	df	AIC	deltaAIC	weight
GAM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + s(\text{time}) + s(\text{fs}, \text{bs} = \text{"re"})$	192,0631354	6419,064422	0	0,159565485
GAM	$Y \sim \text{treatment} + \text{pool} + \text{origin_treatment} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	191,8368416	6419,504578	0,440156558	0,128044278
GAM	$Y \sim s(\text{time}, \text{pool}, \text{bs} = \text{"fs"}) + s(\text{fs}, \text{bs} = \text{"re"})$	192,6318581	6419,862359	0,797937344	0,107070311
GAM	$Y \sim \text{treatment} + \text{pool} + \text{biel} + \text{origin_treatment} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	192,2617842	6419,909574	0,845152596	0,104572238
GAM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + s(\text{fs}, \text{bs} = \text{"re"})$	192,8692376	6420,614425	1,550003272	0,073512302
GAM	$Y \sim \text{pool} + s(\text{time}, \text{by} = \text{pool}) + s(\text{fs}, \text{bs} = \text{"re"})$	192,9872092	6420,697014	1,632592598	0,070538461
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{fs}, \text{bs} = \text{"re"})$	192,9549277	6420,731037	1,666615696	0,069348641
GAM	$Y \sim \text{treatment} + \text{pool} + s(\text{time}, \text{by} = \text{pool}) + s(\text{fs}, \text{bs} = \text{"re"})$	193,0730758	6420,787704	1,723282705	0,067411326
GAM	$Y \sim \text{treatment} + \text{pool} + \text{origin_treatment} + \text{height} + s(\text{time}, \text{by} = \text{treatment}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	192,1487479	6420,880986	1,816564204	0,064339407
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	192,6600833	6420,971494	1,907072231	0,061492689
GAM	$Y \sim \text{treatment} + \text{pool} + \text{temperature_ellerby} + \text{origin_treatment} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	194,0456779	6421,798312	2,733890402	0,040670752
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	187,0778649	6423,39025	4,325828184	0,018348362
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{poisson_fs}, \text{bs} = \text{"re"})$	187,0778649	6423,39025	4,325828184	0,018348362
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{poisson_fs}, \text{bs} = \text{"re"})$	187,0777162	6423,577738	4,513316232	0,016706474
GAM	$Y \sim \text{treatment} + \text{pool} + \text{origin_treatment} + \text{height} + s(\text{time}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	200,2262394	6436,16266	17,09823788	3,09103E-05
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}) + \text{poisson} + \text{fs}$	190,0667063	6578,580453	159,5160313	3,66837E-36
GAM	$Y \sim s(\text{time}) + s(\text{fs}, \text{bs} = \text{"re"})$	10	6804,571181	385,5067594	3,09892E-85
GAMMM	$Y \sim s(\text{time}) + \text{height} + \text{origin_treatment} \mid \text{random} = \text{fs} + \text{poisson}$	7	6810,171014	391,1065923	1,88461E-86
GAMMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) \mid \text{random} = \text{fs}$	8	6813,878664	394,8142421	2,95199E-87
GAMMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{origin_treatment} \mid \text{random} = \text{fs} + \text{poisson}$	12	6815,688152	396,6237299	1,19451E-87
GAMMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{fs}$	9	6817,046094	397,9816722	6,05783E-88
GAMMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{fs} + \text{poisson}$	11	6817,893706	398,8292839	3,96516E-88
GAMMM	$Y \sim s(\text{time}, \text{by} = \text{treatment}) + \text{height} + \text{origin_treatment} \mid \text{random} = \text{fs} + \text{poisson}$	12	6818,123309	399,0588872	3,53511E-88
GAMMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{fs}$	10	6823,544725	404,4803035	2,35048E-89
GAMMM	$Y \sim s(\text{time}) \mid \text{random} = \text{fs}$	5	6959,582138	540,5177164	6,7765E-119
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = \text{"re"})$	80,33673466	7449,552712	1030,48829	2,7244E-225
GAMMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{poisson}$	10	7622,916987	1203,852565	6,1614E-263

Appendix 14: Table of criteria model selection for the acceleration

model_type	formula	df	AIC	deltaAIC	weight
GAM	$Y \sim s(\text{time})$	9,152929265	-16979,15594	0	0,99836855
GAM	$Y \sim s(\text{time}, \text{by} = \text{treatment})$	12,86132149	-16966,32263	12,83330626	0,00163145
GAMM	$Y \sim s(\text{time}, \text{by} = \text{treatment}) \mid \text{random} = \text{none}$	6	-16905,49637	73,65957279	1,00999E-16
GAM	$Y \sim \text{pool} + s(\text{time})$	9,997495323	-16710,6635	268,4924415	4,9762E-59
GAM	$Y \sim \text{treatment} + \text{pool} + s(\text{time})$	10,993335646	-16708,80963	270,3463057	1,96941E-59
GAM	$Y \sim \text{treatment} + \text{height} + \text{pool} + s(\text{time})$	11,99276929	-16707,13121	272,0247281	8,50887E-60
GAM	$Y \sim \text{treatment} + \text{height} + \text{pool} + s(\text{time}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	12,00910992	-16707,10021	272,0557285	8,37799E-60
GAM	$Y \sim \text{treatment} + \text{height} + \text{pool} + s(\text{time}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{poisson_fs}, \text{bs} = \text{"re"})$	12,00910992	-16707,10021	272,0557285	8,37799E-60
GAM	$Y \sim \text{treatment} + \text{height} + \text{pool} + s(\text{time}) + s(\text{poisson}, \text{bs} = \text{"re"})$	12,02346473	-16707,07363	272,0823077	8,26739E-60
GAM	$Y \sim \text{treatment} + \text{pool} + \text{origin_treatment} + \text{height} + s(\text{time}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	12,86942205	-16705,7828	273,3731362	4,33579E-60
GAM	$Y \sim s(\text{time}, \text{by} = \text{pool})$	13,78087317	-16704,80502	274,3509213	2,65916E-60
GAM	$Y \sim \text{treatment} + \text{height} + \text{pool} + \text{origin_treatment} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	17,43635877	-16698,45854	280,6973961	1,11333E-61
GAM	$Y \sim \text{treatment} + \text{height} + \text{pool} + \text{origin_treatment} + s(\text{time}, \text{by} = \text{treatment}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	16,28246988	-16694,27082	284,8851221	1,37174E-62
GAM	$Y \sim \text{treatment} + \text{height} + \text{bief} + \text{pool} + \text{origin_treatment} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	23,76911076	-16686,2888	292,8671398	2,53513E-64
GAM	$Y \sim \text{treatment} + \text{height} + \text{temperature_ellerby} + \text{pool} + \text{origin_treatment} + s(\text{time}, \text{by} = \text{pool}) + s(\text{poisson}, \text{bs} = \text{"re"}) + s(\text{fs}, \text{bs} = \text{"re"})$	52,37426148	-16640,60995	338,5459882	3,05467E-74
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{treatment} + \text{pool} \mid \text{random} = \text{none}$	8	-16608,81361	370,3423329	3,80606E-81
GAMM	$Y \sim s(\text{time}) + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{fs} + \text{poisson}$	9	-16607,16266	371,9932829	1,66715E-81
GAMM	$Y \sim s(\text{time}) + \text{height} + \text{origin_treatment} \mid \text{random} = \text{fs} + \text{poisson}$	10	-16591,74141	387,4145291	7,46956E-85
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{none}$	9	-16585,6281	393,527839	3,51404E-86
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{poisson}$	10	-16583,6281	395,5278398	1,29274E-86
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{height} + \text{origin_treatment} \mid \text{random} = \text{fs} + \text{poisson}$	12	-16566,26211	412,8938307	2,19046E-90
GAMM	$Y \sim s(\text{time}, \text{by} = \text{treatment}) + \text{height} + \text{origin_treatment} \mid \text{random} = \text{fs} + \text{poisson}$	12	-16563,12127	416,0346673	4,55523E-91
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + s(\text{time}) + s(\text{fs}, \text{bs} = \text{"re"})$	199,6130544	-16365,64794	613,5079998	5,9941E-134
GAM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + s(\text{time}) + s(\text{fs}, \text{bs} = \text{"re"})$	192,0631354	6419,064422	23398,22036	0
GAM	$Y \sim s(\text{time}, \text{pool}, \text{bs} = \text{"fs"}) + s(\text{fs}, \text{bs} = \text{"re"})$	192,6318581	6419,862359	23399,0183	0
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) + \text{pool} \mid \text{random} = \text{none}$	7	311027,5703	328006,7263	0
GAMM	$Y \sim s(\text{time}, \text{by} = \text{pool}) \mid \text{random} = \text{none}$	6	311252,8052	328231,9611	0
GAMM	$Y \sim s(\text{time}) \mid \text{random} = \text{none}$	4	316552,1355	333531,2914	0

Appendix 15: Temporal dynamics of the cumulated difference of angle per segment.

We modeled the temporal trajectories for each segment, being rather similar (to be tested, different contribution)



Appendix 16: Table of criteria model selection for cumulated angle difference per segment

model_type	formula	df	AIC	deltaAIC	weight
GAM	$Y \sim \text{treatment} + \text{pool} + \text{origin_treatment} + \text{height} + \text{s(time, by = pool)} + \text{segment} + \text{s(poisson, bs = "re") + s(fs, bs = "re")}$	213,1082838	295418,0014	0	0,152377492
GAM	$Y \sim \text{treatment} + \text{pool} + \text{bief} + \text{origin_treatment} + \text{height} + \text{s(time, by = pool)} + \text{segment} + \text{s(poisson, bs = "re") + s(fs, bs = "re")}$	213,2014883	295418,0703	0,068964514	0,14721273
GAM	$Y \sim \text{pool} + \text{height} + \text{segment} + \text{s(time, by = pool)} + \text{s(fs, bs = "re")}$	213,2092297	295418,0713	0,069925409	0,147142019
GAM	$Y \sim \text{height} + \text{segment} + \text{s(time, by = pool)} + \text{s(fs, bs = "re")}$	213,2167333	295418,0716	0,070266426	0,147116932
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + \text{segment} + \text{s(time, by = pool)} + \text{s(fs, bs = "re")}$	213,2732311	295418,1434	0,142016358	0,141932666
GAM	$Y \sim \text{height} + \text{segment} + \text{s(time, pool, bs = "fs") + s(fs, bs = "re")}$	213,3611179	295418,6762	0,674826988	0,108738666
GAM	$Y \sim \text{segment} + \text{s(time, by = pool)} + \text{s(fs, bs = "re")}$	214,1404242	295419,1505	1,149150103	0,085780001
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + \text{segment} + \text{s(time, by = pool)} + \text{s(poisson, bs = "re") + s(fs, bs = "re")}$	214,3689166	295420,507	2,505641447	0,043533912
GAM	$Y \sim \text{treatment} + \text{pool} + \text{temperature_ellery} + \text{origin_treatment} + \text{height} + \text{s(time, by = pool)} + \text{segment} + \text{s(poisson, bs = "re") + s(fs, bs = "re")}$	215,6548962	295421,9366	3,935227175	0,021300858
GAM	$Y \sim \text{height} + \text{segment} + \text{s(time, by = pool)} + \text{s(fs, bs = "re")}$	215,4748528	295424,8901	6,88870153	0,004864725
GAM	$Y \sim \text{treatment} + \text{pool} + \text{origin_treatment} + \text{height} + \text{s(time, by = treatment)} + \text{segment} + \text{s(poisson, bs = "re") + s(fs, bs = "re")}$	213,0545312	295644,839	226,837635	8,42858E-51
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + \text{s(time)} + \text{segment} + \text{s(poisson, bs = "re") + s(fs, bs = "re")}$	207,2645784	295653,4277	235,4263033	1,15013E-52
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + \text{s(time)} + \text{segment} + \text{s(poisson, bs = "re") + s(fs, bs = "re")}$	207,2645784	295653,4277	235,4263033	1,15013E-52
GAM	$Y \sim \text{treatment} + \text{pool} + \text{origin_treatment} + \text{height} + \text{s(time)} + \text{segment} + \text{s(poisson, bs = "re") + s(fs, bs = "re")}$	207,5698356	295654,0508	236,0494356	8,42242E-53
GAM	$Y \sim \text{treatment} + \text{pool} + \text{height} + \text{s(time)} + \text{poisson} + \text{fs} + \text{segment}$	212,3279258	295657,5594	239,5580169	1,45733E-53
GAMM	$Y \sim \text{segment} + \text{s(time, by = pool)} + \text{height} + \text{origin_treatment} \mid \text{random} = \text{fs} + \text{poisson}$	23	295916,6028	498,6014334	8,1846E-110
GAMM	$Y \sim \text{segment} + \text{s(time, by = pool)} + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{fs} + \text{poisson}$	22	295922,5785	504,5770739	4,1248E-111
GAMM	$Y \sim \text{segment} + \text{s(time, by = pool)} + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{fs} + \text{poisson}$	21	295923,0976	505,0962546	3,1817E-111
GAMM	$Y \sim \text{segment} + \text{s(time, by = pool)} + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{fs}$	20	295925,0278	507,0263769	1,2121E-111
GAMM	$Y \sim \text{segment} + \text{s(time, by = pool)} + \text{height} \mid \text{random} = \text{fs}$	19	295929,8398	511,8384544	1,093E-112
GAMM	$Y \sim \text{segment} + \text{s(time, by = pool)} + \text{height} \mid \text{random} = \text{fs}$	18	295957,7464	539,7450539	9,523E-119
GAMM	$Y \sim \text{segment} + \text{s(time, by = pool)} \mid \text{random} = \text{fs}$	23	296139,7926	721,7912628	2,805E-158
GAM	$Y \sim \text{segment} + \text{s(time, by = treatment)} + \text{height} + \text{origin_treatment} \mid \text{random} = \text{fs} + \text{poisson}$	97,50174792	298347,6606	2929,659236	0
GAMM	$Y \sim \text{treatment} + \text{pool} + \text{height} + \text{segment} + \text{s(time, by = pool)} + \text{s(poisson, bs = "re")}$	21	298568,654	3150,652613	0
GAM	$Y \sim \text{segment} + \text{s(time, by = pool)} + \text{height} + \text{treatment} + \text{pool} \mid \text{random} = \text{poisson}$	209,8530527	300376,8279	4958,826515	0
GAMM	$Y \sim \text{segment} + \text{s(time)} \mid \text{random} = \text{fs}$	16	300898,1535	5480,15213	0
GAM	$Y \sim \text{s(time, by = pool)} + \text{s(fs, bs = "re")}$	200,0892678	307256,5046	11838,50321	0
GAMM	$Y \sim \text{s(time, by = pool)} \mid \text{random} = \text{fs}$	7	307730,9999	12312,99856	0

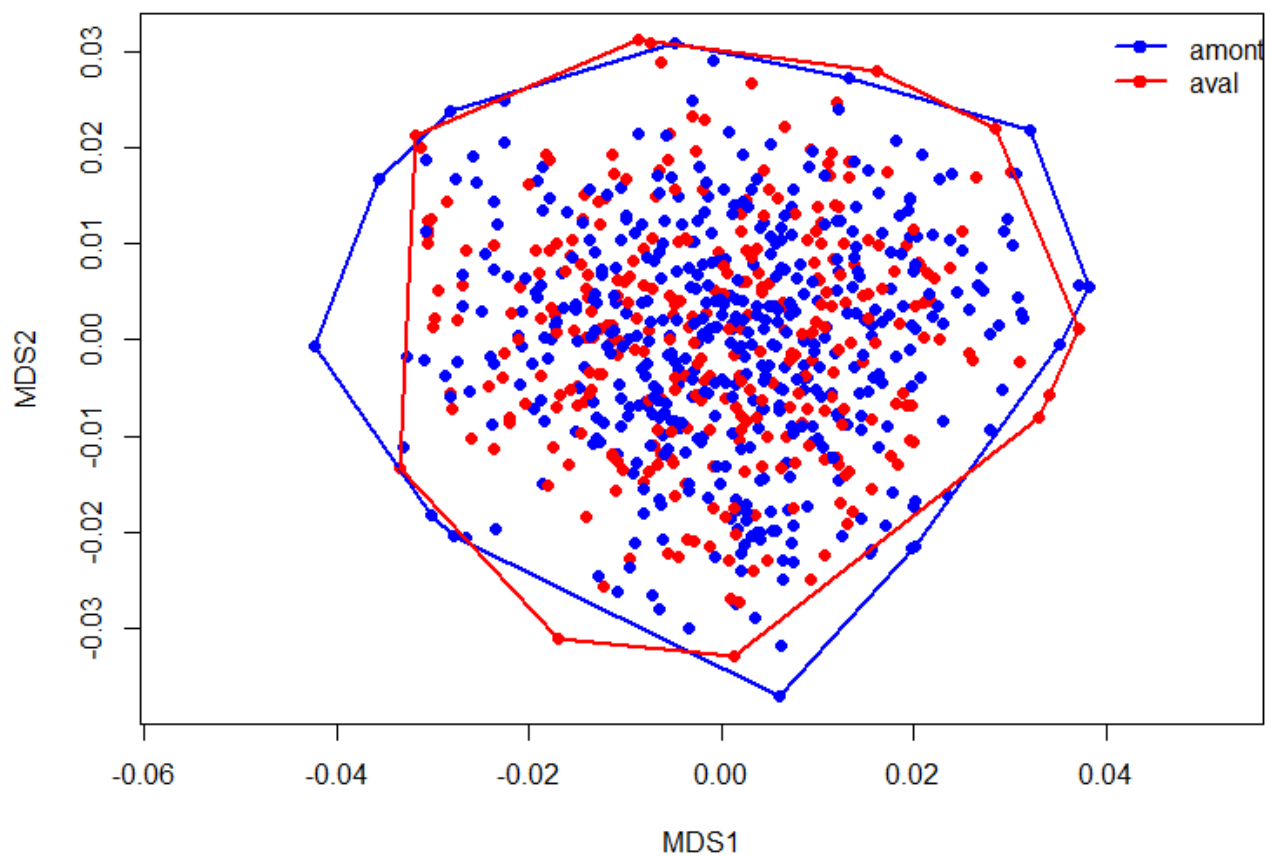
Appendix 17: Table of criteria model selection for the evolution snout-tail ratio

model_type	formula	df	AIC	deltaAIC	weight
GAM	absY ~ s(time, by = pool) + s(time) + s(fs, bs = "re")	130,9349129	24198,34459	0	0,363967522
GAM	absY ~ s(time, by = pool) + s(fs, bs = "re")	131,1482212	24199,39336	1,048768558	0,215439546
GAM	absY ~ s(time, by = pool) + treatment + pool + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	131,8324339	24200,4026	2,058016092	0,130067896
GAM	absY ~ s(time, pool, bs = "fs") + s(fs, bs = "re")	131,2683917	24201,7405	3,395912013	0,066626916
GAM	absY ~ s(time, by = pool) + bief + s(fs, bs = "re")	133,4449299	24202,0727	3,728115962	0,056430323
GAM	absY ~ s(time, by = pool) + bief + height + s(poisson, bs = "re") + s(fs, bs = "re")	133,8899944	24202,43738	4,092797347	0,047024369
GAM	absY ~ s(time, by = pool) + bief + s(poisson, bs = "re") + s(fs, bs = "re")	133,7051141	24202,59177	4,247183194	0,043530988
GAM	absY ~ s(time, by = pool) + pool + bief + height + s(poisson, bs = "re") + s(fs, bs = "re")	134,7262327	24203,33538	4,99078751	0,030014208
GAM	absY ~ s(time, by = pool) + treatment + pool + bief + height + s(poisson, bs = "re") + s(fs, bs = "re")	134,8167669	24203,57444	5,229853532	0,026632649
GAM	absY ~ s(time, by = pool) + pool + bief + treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	135,4828944	24204,15997	5,815384651	0,019873204
GAM	absY ~ s(time, by = pool) + treatment + pool + temperature_elleryby + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	142,6166442	24212,75221	14,40762511	0,000270699
GAM	absY ~ s(time, by = pool) + treatment + bief + temperature_elleryby + pool + bief + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	144,4300557	24214,35146	16,00687436	0,000121679
GAM	absY ~ s(time, by = height) + treatment + pool + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	120,6271998	24274,96192	76,61733437	8,39112E-18
GAM	absY ~ s(time, by = treatment) + treatment + pool + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	128,6185439	24280,6255	82,2809119	4,94299E-19
GAM	absY ~ s(time) + treatment + pool + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	120,7715341	24281,7748	83,43021598	2,78241E-19
GAM	absY ~ s(time) + treatment + pool + height + s(poisson, bs = "re") + s(fs, bs = "re")	121,9670219	24283,49556	85,1509734	1,17697E-19
GAMM	absY ~ s(time, by = pool) + height + treatment + pool + bief random = fs + poisson	17	24288,76605	90,42146499	8,439E-21
GAMM	absY ~ s(time, by = pool) + bief random = fs	13	24290,87817	92,53358713	2,93528E-21
GAMM	absY ~ s(time, by = pool) + bief random = fs + poisson	14	24292,87818	94,53359239	1,07983E-21
GAMM	absY ~ s(time, by = pool) + height + pool + bief random = fs + poisson	16	24294,29856	95,95397001	5,3079E-22
GAMM	absY ~ s(time, by = pool) + height + bief random = fs + poisson	15	24296,8427	98,49811432	1,48754E-22
GAMM	absY ~ s(time, by = pool) + height + origin_treatment random = fs + poisson	12	24307,16966	108,8250707	8,51136E-25
GAM	absY ~ s(time, by = pool) + bief + s(poisson, bs = "re")	62,65564687	24309,5748	111,2302107	2,55699E-25
GAMM	absY ~ s(time, by = pool) random = fs	7	24312,87005	114,525467	4,92236E-26
GAMM	absY ~ s(time, by = pool) + bief random = poisson	13	24357,65918	159,3145913	9,2542E-36
GAM	absY ~ s(time) + treatment + pool + height + poisson + fs	201,837433	24362,07314	163,7285523	1,01826E-36
GAMM	absY ~ s(time) + height + origin_treatment random = fs + poisson	10	24363,18703	164,8421391	5,83421E-37
GAMM	absY ~ s(time, by = treatment) + height + origin_treatment random = fs + poisson	12	24377,95924	179,6146503	3,61607E-40
GAM	absY ~ s(time) + bief + s(fs, bs = "re")	123,4489634	24685,49	487,1454111	6,0087E-107
GAMM	absY ~ s(time) + bief random = fs	11	24748,09619	549,7516053	1,5276E-120

Appendix 18: Table of criteria model selection for the evolution of the anterior-posterior ratio

model_type	formula	df	AIC	deltaAIC	weight
GAM	absY ~ s(time, by = pool) + s(fs, bs = "re")	136,1579933	22663,02414	0	0,441522095
GAM	absY ~ s(time, pool, bs = "fs") + s(fs, bs = "re")	135,7222244	22663,7943	0,770157855	0,300410279
GAM	absY ~ s(time, by = pool) + bief + s(fs, bs = "re")	137,8722303	22665,2836	2,259452615	0,142665361
GAM	absY ~ s(time, by = pool) + s(time) + s(fs, bs = "re")	137,8771519	22666,85609	3,831944943	0,064991442
GAM	absY ~ s(time, by = pool) + treatment + pool + bief + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	139,5193885	22667,45872	4,43457498	0,048083572
GAM	absY ~ s(time, by = pool) + treatment + pool + temperature_ellertby + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	146,2424113	22674,45555	11,43140884	0,001454288
GAM	absY ~ s(time, by = pool) + treatment + bief + temperature_ellertby + pool + bief + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	147,5608025	22675,49899	12,47484206	0,000863127
GAM	absY ~ s(time, by = pool) + pool + bief + height + s(poisson, bs = "re") + s(fs, bs = "re")	148,0439207	22685,58706	22,56291968	5,56515E-06
GAM	absY ~ s(time, by = pool) + treatment + pool + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	150,041579	22688,38607	25,36192956	1,37303E-06
GAM	absY ~ s(time, by = pool) + pool + bief + treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	149,0799415	22688,61246	25,58831982	1,22608E-06
GAM	absY ~ s(time, by = pool) + pool + bief + height + s(poisson, bs = "re") + s(fs, bs = "re")	150,678123	22689,0497	26,02555949	9,85313E-07
GAM	absY ~ s(time, by = pool) + bief + height + s(poisson, bs = "re") + s(fs, bs = "re")	150,3802784	22689,80187	26,7777299	6,7646E-07
GAM	absY ~ s(time, by = pool) + treatment + pool + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	127,1063874	22720,12119	57,09704829	1,76395E-13
GAM	absY ~ s(time, by = height) + treatment + pool + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	126,6561472	22723,76472	60,74057323	2,85302E-14
GAM	absY ~ s(time, by = treatment) + treatment + pool + origin_treatment + height + s(poisson, bs = "re") + s(fs, bs = "re")	146,1597114	22742,65634	79,63219224	2,25446E-18
GAM	absY ~ s(time) + treatment + pool + height + s(poisson, bs = "re") + s(fs, bs = "re")	139,0390582	22743,34198	80,31783136	1,60014E-18
GAMM	absY ~ s(time, by = pool) + bief random = fs + poisson	17	22765,11064	102,086492	3,00021E-23
GAMM	absY ~ s(time, by = pool) + bief random = fs	13	22766,43087	103,4067279	1,55048E-23
GAMM	absY ~ s(time, by = pool) + bief random = fs + poisson	14	22768,2229	105,198754	6,32897E-24
GAMM	absY ~ s(time, by = pool) + height + pool + bief random = fs + poisson	16	22770,23366	107,2095206	2,3158E-24
GAMM	absY ~ s(time, by = pool) + height + bief random = fs + poisson	15	22772,24566	109,2215159	8,4684E-25
GAM	absY ~ s(time, by = pool) + bief + s(poisson, bs = "re")	65,98378101	22777,5897	114,5655529	5,85273E-26
GAMM	absY ~ s(time, by = pool) + height + origin_treatment random = fs + poisson	12	22781,61019	118,5860449	7,84007E-27
GAMM	absY ~ s(time, by = pool) random = fs	7	22786,05343	123,029286	8,50124E-28
GAM	absY ~ s(time) + treatment + pool + height + poisson + fs	201,7521395	22794,20905	131,1849017	1,4405E-29
GAMM	absY ~ s(time) + height + origin_treatment random = fs + poisson	10	22811,83289	148,8087445	2,14558E-33
GAMM	absY ~ s(time, by = treatment) + height + origin_treatment random = fs + poisson	12	22827,85965	164,8355106	7,10193E-37
GAMM	absY ~ s(time, by = pool) + bief random = poisson	13	22836,86872	173,844573	7,85386E-39
GAM	absY ~ s(time) + bief + s(fs, bs = "re")	128,8921054	23099,31516	436,2910167	8,04566E-36
GAMM	absY ~ s(time) + bief random = fs	11	23174,57471	511,5505623	3,6573E-112

Appendix 19: Illustration of the morphospace depending on the treatment conditions (stress=0.178). Adapted from M. Descat.



8. Supplementary data

All data and protocols for statistical analysis, modelling and numeric animation are available via [Github](#). Contact: felix.guillaud.work@gmail.com